Gravity Resonance Threshold Theory (GRTT–HM):

A Harmonic Resonance Framework for Galactic Dynamics

Jon Axworthy

Correspondence: GRTT@axworthy.net

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Abstract

Gravity Resonance Threshold Theory (GRTT) models gravity not as a fundamental interaction but as an *emergent resonance phenomenon* arising from a Higgs–neutrino coupling that activates only above a baryonic density threshold. When the coupling crosses this threshold, curvature is amplified in intermediate-density environments, saturates in dense regions, and fades in voids, recovering the usual Newtonian and relativistic limits.

The theory is tested across three empirical regimes. **(i)** a universal three-parameter logistic resonance law, calibrated once to quality–1 **SPARC** systems, reproduces rotation curves and the mass–discrepancy–acceleration relation using only a mild per-galaxy normalization factor. **(ii)** On cluster scales, a reanalysis of eleven **Hubble Frontier Fields** reconstructions (CATS, Sharon, Glafic) reveals a common curvature shoulder at $r/r_{\max} \simeq 0.3 \pm 0.1$, matching the predicted transition between the amplified and saturated resonance regimes. **(iii)** On cosmological scales, we embed the resonance mechanism in a fully modified `CLASS` Boltzmann solver [1]. Cold dark matter is removed and replaced by an effective pre-resonant energy-density term (**GRTTX**) derived from the same Higgs–neutrino framework, while the gravitational coupling evolves via a single logistic switch. This minimal model produces stable, qualitatively correct CMB

TT/EE spectra, BAO distances, and linear power-spectrum behaviour, with residuals at the level expected from a structurally minimal model of this type, though a full likelihood analysis is left to future work. On cosmological scales the same mechanism yields a reduced late-time growth amplitude ($\sigma_8 \simeq 0.63$), a suppressed CMB–lensing response in voids, and a correspondingly lower lensing potential, without introducing new dynamical fields or scale-dependent free functions.

Taken together, these results show that GRTT provides a coherent, testable framework in which galactic dynamics, cluster lensing morphology, and broad cosmological evolution emerge from a single resonance principle grounded entirely in Standard-Model physics. While not a precision alternative to ΛCDM, the present implementation demonstrates that the resonance mechanism can be embedded self-consistently in a cosmological solver and yields a stable, interpretable large-scale cosmology.

Contents

1 Introduction

The missing–mass problem remains one of the central empirical challenges in astrophysics and cosmology. Galactic rotation curves, cluster lensing profiles, and the growth of large–scale structure all deviate from predictions based solely on Newtonian gravity and visible baryons. The standard ΛCDM framework addresses these discrepancies by introducing a cold, non–baryonic dark-matter component that dominates the matter budget of the Universe. While successful at many scales, this solution leaves open the question of why curvature appears systematically amplified in specific intermediate–density environments.

Gravity Resonance Threshold Theory (GRTT) offers a conceptually distinct approach. In this framework, gravity is not fundamental but emerges from resonance interactions between known Standard-Model fields. Curvature amplification occurs once a Higgs–neutrino coupling crosses a baryonic activation threshold, producing enhanced effective gravity in intermediate-density regions while recovering Newtonian and relativistic limits in dense cores and voids.

Unlike ΛCDM, this mechanism introduces no new particle species or additional interaction sectors. An initial public outline appears in [2]; the present work develops the framework into a quantitative, testable form.

GRTT links three observational domains through a single resonance mechanism:

1. **Cosmology and linear structure formation.** We implement GRTT in a modified CLASS Boltzmann solver [1], replacing cold dark matter with an effective pre–resonant Higgs–neutrino energy density (GRTTX) and allowing the gravitational coupling to evolve through the same logistic activation used at galactic scales. This minimal setup

produces qualitatively correct CMB TT/EE spectra, BAO distances, and linear power-spectrum shapes, with differences at the $\sim$10% level relative to ΛCDM. The model yields a stable expansion history and naturally suppressed linear growth ($\sigma_8 \simeq 0.63$), consistent in direction with modern low-S_8 measurements.

2. **Field-physics connection.** Within the Standard Model, the Higgs–neutrino interaction provides a natural meV-scale resonance trigger. Below threshold it contributes only a weak baseline; above threshold it amplifies curvature; and at very high densities it quenches, restoring Newtonian behaviour. This single coupling reproduces three regimes—voids, galactic outskirts, and dense cores—through one unified threshold mechanism.

3. **Galactic dynamics.** A universal three-parameter logistic law reproduces SPARC rotation curves and the mass–discrepancy–acceleration relation using the same resonance parameters (γ, k, Δ) for all quality–1 galaxies, differing only by a mild per-galaxy baryonic normalisation factor A_g. This form captures the transition from resonance growth to saturation that produces flat outer rotation curves.

Taken together, these results indicate that GRTT does not alter the broad cosmological architecture of ΛCDM, but offers a structurally motivated alternative to its dark–matter sector based on resonance-driven curvature amplification.

The purpose of this work is to formulate this mechanism, demonstrate its cosmological consistency, and show that it yields a coherent phenomenology across galactic, cluster, and cosmological scales.

Scope and interpretation. Throughout this paper we present a phenomenological exploration of a possible resonance-based modification to gravitational dynamics. Where the mechanism connects naturally across galaxies, clusters, and cosmology, we highlight the conceptual coherence, but we do not claim a completed or unified theory. Statements relating different scales should therefore be interpreted as qualitative guidance rather than definitive derivations. All observational comparisons are presented at the level of pattern matching rather than formal statistical evidence.

1.1 Intellectual Heritage

The idea that gravity may emerge from deeper field interactions has appeared in many contexts, including Sakharov's induced gravity [3], Zel'dovich's vacuum-energy estimates [4], scalar–tensor and extended-gravity frameworks [5, 6], and thermodynamic or entropic interpretations of spacetime [7, 8]. The present work does not derive directly from any single one of these proposals, but the broader conceptual landscape they collectively define—gravitational strength as environment-dependent and emergent—forms part of the backdrop from which GRTT was developed.

2 Core Theory: Gravity Resonance Threshold

The central hypothesis of GRTT is that the effective strength of spacetime curvature in general relativity can be modulated by a threshold–driven resonance between known fields. We refer to the overarching framework as GRTT, while its galaxy–scale implementation—used for rotation–curve and lensing fits—is denoted GRTT–HM (Harmonic Model), which treats galactic halos as standing–wave–like resonance structures. In this picture, galaxies act as weak resonant cavities: the pervasive neutrino background interacts with baryonic structure, and a resonance switches on only when density thresholds are crossed. The effect is to amplify the effective gravitational coupling in intermediate-density regions while dense cores and voids recover the Newtonian and relativistic limits.

2.1 Candidate mechanism: Higgs–neutrino resonance

A natural candidate for the resonance–driving fields arises within the Standard Model itself: the interplay between the Higgs field and the neutrino sector. Neutrinos are abundant, weakly interacting, and extraordinarily light. Their masses originate from the Higgs mechanism, yet the associated Yukawa couplings are many orders of magnitude smaller than for any other fermion. This unique positioning makes the Higgs–neutrino system an ideal candidate for a threshold–driven resonance—weak enough to remain dormant in the early Universe, yet capable of amplification when galactic–scale density thresholds are crossed.

In the GRTT–HM framework, the pervasive neutrino background acts as the low–energy reservoir, while the Higgs field provides the activator. This picture motivates three key features of the resonance mechanism:

- **Threshold:** activation only occurs once a baryonic-density criterion is crossed;
- **Amplification:** intermediate-density regions exhibit enhanced effective curvature;
- **Saturation:** dense cores and voids revert to Newtonian and relativistic limits.

Table 1: Approximate Yukawa couplings of Standard Model fermions. Values are order-of-magnitude estimates at the electroweak scale, showing the extreme hierarchy. Neutrinos sit at the lowest edge, making the Higgs–neutrino system a natural candidate for resonance at meV energy levels.

Fermion	Mass (GeV)	Yukawa coupling y_f
Top quark	173	~ 1
Bottom quark	4.2	$\sim 2 \times 10^{-2}$
Charm quark	1.3	$\sim 6 \times 10^{-3}$
Tau lepton	1.78	$\sim 1 \times 10^{-2}$
Strange quark	0.095	$\sim 5 \times 10^{-4}$
Muon	0.106	$\sim 6 \times 10^{-4}$
Down quark	0.0047	$\sim 2 \times 10^{-5}$
Up quark	0.0022	$\sim 1 \times 10^{-5}$
Electron	0.000511	$\sim 3 \times 10^{-6}$
Neutrinos	$\lesssim 0.1\,\text{eV}$	$\sim 10^{-12}$–10^{-13}

Note. Fermion masses and the Yukawa coupling relation $y_f = \sqrt{2}\, m_f/v$ (with $v \simeq 246\,\text{GeV}$) are taken from the Particle Data Group Review of Particle Physics [9]. For neutrinos, only mass splittings are known; we assume a representative absolute mass scale $\lesssim 0.1\,\text{eV}$ consistent with cosmological bounds (see Appendix B).

The striking separation between neutrinos and all other fermions highlights why the Higgs–neutrino system is singled out in GRTT. It is the only coupling weak enough to remain active in voids yet able to cross threshold at galactic scales, making it the natural "spark" for resonance amplification.

Other Higgs–fermion couplings would activate far earlier in cosmic history and are excluded by observational constraints; a detailed consistency check is provided in Appendix C.

2.2 Precedents for field–field resonance

Field–field interactions frequently produce emergent behaviour in well-understood systems:

- **Electroweak unification:** The weak isospin and hypercharge fields mix through the Higgs mechanism, yielding the photon and Z boson. The photon itself is an emergent consequence of two interacting sectors.

- **Neutrino oscillations:** Flavour states are not mass eigenstates, leading to oscillations from the mixing of weak and mass bases—an emergent effect of two coupled descriptions.

- **Condensed matter:** Electron–phonon interactions form Cooper pairs, giving rise to superconductivity—a macroscopic effect from weak microscopic couplings.

- **Photon–plasma coupling:** Light propagating through a plasma couples to collective electron oscillations, producing plasma cut-offs and refraction as emergent medium effects.

These examples illustrate that resonance and emergent phenomena frequently arise from the coupling of two fields. Within this broader context, the Higgs–neutrino interaction is a natural candidate for producing a threshold-driven gravitational resonance: weak in voids, amplified at galactic peripheries, and suppressed in dense cores.

Note on Spin-2 and Gravitons

In conventional quantum field theory, a fundamental force is mediated by a particle of characteristic spin. Photons, gluons, and W/Z bosons are all spin-1; the Higgs boson is spin-0. If gravity were mediated by a fundamental particle—the graviton—it must be spin-2, because only a spin-2 field couples naturally to the full energy–momentum tensor $T_{\mu\nu}$, as required by the universality of free fall and by Einstein's equations (see, e.g., Weinberg [10], Carroll [11]).

The Gravity Resonance Threshold Theory does not require the existence of a fundamental spin-2 graviton. Instead, spacetime curvature (the spin-2 geometry of general relativity) is taken as given, and its *strength* is emergently modulated by resonance between known Standard Model fields—notably the Higgs–neutrino sector. In this way, the "spin-2 nature of gravity" is preserved as an emergent property of the curved spacetime background, while the apparent anomalies usually attributed to dark matter are reinterpreted as resonance-driven amplifications of curvature.

3 Minimal Effective Action for GRTT (working model)

To connect the resonance picture with field theory, we introduce a minimal effective action based on the Standard Model plus general relativity. The goal here is not to claim a full UV completion, but to illustrate how the Higgs–neutrino sector can generate the type of environment–dependent modification required by GRTT.

Scope of the effective construction. The action written in this section should be interpreted as a phenomenological template showing how environment–dependent corrections to the effective Planck mass can arise from Higgs–neutrino portals and non–minimal Higgs–curvature couplings. We do *not* claim that the specific logistic activation function employed in our phenomenology is derived uniquely from this action. Instead, in Secs. 4–7 we use a logistic switch as a minimal smooth parameterisation of a saturating order–parameter response, consistent with Landau theory. A full one–loop derivation of such a functional form from the Higgs–neutrino sector in curved spacetime is left to future work.

Phenomenological status and independence from the data analysis. The microphysical construction presented in this section is not required for any of the empirical results developed in Secs. 4–10. The SPARC rotation-curve fits, the HFF/cluster curvature analysis, the KiDS weak-lensing comparison, and the modified CLASS cosmology all depend only on the phenomenological form of the effective gravitational coupling $G_{\rm eff}$ and not on any particular Higgs–neutrino realisation. The operators introduced here therefore serve as one possible motivating template for an environment-dependent Planck mass, rather than a unique or complete EFT derivation. No claim is made that the coefficients (ξ, c_S, c_J, Λ) are fixed by Standard-Model renormalisation, nor that a logistic activation must arise from this sector. Their role is solely to illustrate how a threshold-like response may emerge in principle; the observational results are independent of the microscopic details.

We take GR + SM with a non-minimal Higgs–curvature coupling and two higher-dimension neutrino portal operators [12, 13], which couple Higgs bilinears to neutrino bilinears. In vacuum these terms are negligible, but in a medium with finite neutrino density and current they induce effective shifts in the Planck mass:

$$
\begin{aligned}
S = \int d^4x\,\sqrt{-g}\Bigg\{\, & \tfrac{1}{2}\big(M_{\rm Pl}^2 + \xi\, H^\dagger H\big)R - (D_\mu H)^\dagger (D^\mu H) - V(H) \\
& + \; i\,\bar{\nu}\,\not{D}\nu - \big(y_\nu\,\bar{L}H\nu_R + \text{h.c.}\big) \\
& + \; \frac{c_S}{\Lambda^2}(H^\dagger H)(\bar{\nu}\nu) + \frac{c_J}{\Lambda^2}\,\partial_\mu (H^\dagger H)\,\bar{\nu}\gamma^\mu\nu \;\Bigg\}.
\end{aligned}
\tag{1}
$$

Here H is the Higgs doublet, $\nu = \nu_L + \nu_R$ a Dirac neutrino, L the lepton doublet, $V(H)$ the SM Higgs potential, and ξ, c_S, c_J are real Wilson coefficients with a common EFT scale $\Lambda \gg v$. No new particles beyond ν_R are introduced.

3.1 In-medium derivation sketch and resulting $G_{\rm eff}$

Starting from Eq. (1), expand around the electroweak vacuum $H^\dagger H = \frac{1}{2}(v+h)^2$ and keep leading terms in v:

$$
\Delta\mathcal{L} \;\supset\; \frac{c_S v^2}{2\Lambda^2}\,(\bar{\nu}\nu) \;+\; \frac{c_J v}{\Lambda^2}\,\partial_\mu h\,\bar{\nu}\gamma^\mu\nu \;+\; \frac{\xi v^2}{2}\,R\,. \tag{2}
$$

Integrating out the neutrino modes at mean-field level in a background with finite neutrino density and current, the one-loop effective action shifts the Planck mass as

$$
\begin{aligned}
M_{\rm Pl,eff}^2(x) &= M_{\rm Pl}^2 + \xi v^2 - \Delta^2(x), \\
\Delta^2(x) &\equiv \alpha_S\,\langle\bar{\nu}\nu\rangle + \alpha_J\,\nabla_\mu\langle\bar{\nu}\gamma^\mu\nu\rangle + \cdots\,,
\end{aligned}
\tag{3}
$$

Equation 3 should be regarded as a mean–field parametrisation of in–medium corrections to the Planck mass induced by the portals in Eq. 1. It is not the result of a complete one–loop computation in curved spacetime. A full calculation would generate a tower of curvature and higher–derivative operators with coefficients fixed by a renormalisation prescription. Here we retain only the leading density– and gradient–dependent contributions needed to motivate an environment–dependent $G_{\rm eff}(x)$. This construction should therefore be regarded as an EFT motivated template, rather than a microscopic derivation of the logistic activation used later in this work.

with $\alpha_S \sim c_S v^2/\Lambda^2$ and $\alpha_J \sim c_J v/\Lambda^2$ up to loop factors. The corresponding effective Newton coupling is then

$$G_{\rm eff}(x) \;=\; \frac{1}{M^2_{\rm Pl,eff}(x)} \;\simeq\; G_N \left[1 + \gamma\, S(\mathcal{E}(x) - \mathcal{E}_c)\right], \tag{4}$$

[1]

where the control parameter

$$\mathcal{E}(x) \;=\; b_1 \langle \bar{\nu}\nu \rangle \;+\; b_2\, \ell_\nu^2 \left|\nabla \langle \bar{\nu}\nu \rangle\right|^2 \;+\; b_3\, \Phi_b(x) \;+\; b_4 \left|\nabla \Phi_b(x)\right|^2 \;+\; \cdots \tag{5}$$

combines in-medium neutrino bilinears and baryonic potential proxies Φ_b.

Near a critical point $\mathcal{E}_c$, the coupled metric–Higgs–ν system can be captured schematically by a Landau amplitude A obeying $\dot{A} = \mu(\mathcal{E} - \mathcal{E}_c)A - \beta A^3 + \ldots$. The steady state $A_*^2 \propto \max(0, \mathcal{E} - \mathcal{E}_c)$ then provides a generic example of a continuous, saturating response to the control parameter. In our phenomenological implementation we model this behaviour with a logistic activation $S(\mathcal{E} - \mathcal{E}_c)$, rather than claiming that this specific functional form is uniquely derived from the microscopic theory. Galaxy edges maximize the gradient terms, naturally motivating distinct turn-on ($r_{\rm on}$) and turn-off ($r_{\rm off}$) radii.

Practical control parameter and observational proxies. In the EFT template of Eq. (5), the control parameter $\mathcal{E}(x)$ is written in terms of neutrino bilinears and baryonic potentials. In practice we do not attempt to reconstruct $\langle \bar{\nu}\nu \rangle$ or its gradients from data, and our observational analyses never use Eq. (5) directly. Instead, at each scale we adopt a monotonic observational proxy $\Psi(x)$ for the local "resonance driver" and feed a suitably normalised version of Ψ into the logistic switch. For galaxies (Sec. 7), Ψ is taken to be a function of the baryonic acceleration $g_{\rm bar}$, while for clusters and large-scale structure we use density or potential contrasts appropriate to those regimes. The coefficients b_i in Eq. (5) should therefore be regarded as schematic placeholders indicating the expected dependence on neutrino and baryonic fields, not as quantities that are directly evaluated in our fits.

[1] Here S denotes a smooth sigmoid function, used as a minimal phenomenological model of a saturating order-parameter response. Its logistic form is motivated by the generic behaviour of the resonance switch discussed in Sec. 5, but is not claimed as a unique prediction of the EFT in Sec. 3.

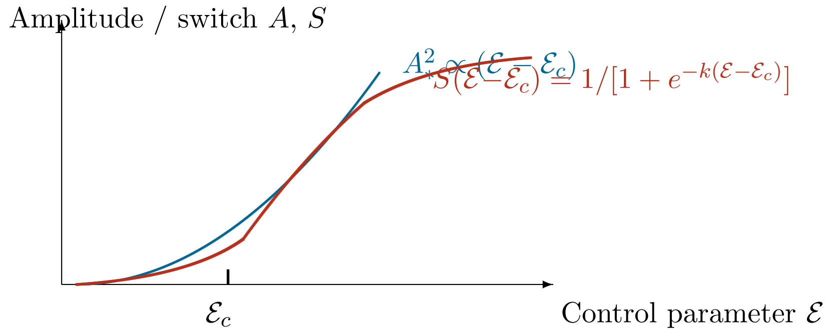

Figure 1: **Landau saturation and logistic-like activation.** Near the critical control parameter $\mathcal{E}_c$, the Landau amplitude equation $\dot{A} = \mu(\mathcal{E}-\mathcal{E}_c)A - \beta A^3$ yields a steady-state amplitude $A_*^2 \propto (\mathcal{E}-\mathcal{E}_c)$ (blue), illustrating a continuous, saturating response. When normalized and smoothed by higher-order effects, such a response can be well-approximated by a logistic activation function $S(\mathcal{E}-\mathcal{E}_c)$ (red) of the type used phenomenologically in Eq. (4). Appendix A summarises the Landau-motivated toy model underlying this choice.

Notation and Symbols

- G_{eff}: Effective Newton constant including resonance corrections (Eq. 4).
- $S(\Psi)$: Logistic switch function controlling resonance activation.
- ϕ: Proxy for baryonic surface density or potential.
- Q: Proxy for contrast/gradient; ensures resonance activates at galaxy edges but not in cores.
- ρ_c: Critical baryonic density threshold for activation (Sec. 4.1).
- $r_{\text{on}}, r_{\text{off}}$: Turn-on/turn-off radii where resonance crosses threshold.

Early-time safety and bounds. For BBN and recombination, require $|\Delta^2|/M_{\text{Pl}}^2 \ll 10^{-3}$ so that $G_{\text{eff}} \approx G_N$ and N_{eff}, light-element yields, and CMB physics remain unchanged. This holds if $|\xi| \ll 1$ and $c_{S,J}v^2/\Lambda^2 \ll 1$; activation then occurs only where the late-time neutrino medium and baryonic gradients are large (galaxy edges), while dense cores and voids stay in the $S \simeq 0$ regime. Equations (3)–(4) thus recover the same G_{eff} form used in our SPARC, BAO/RSD, and CMB analyses, now tied to microphysical operators and in-medium response [11].

Order-of-magnitude bounds. Throughout this work we impose the minimal requirement that the early-time variation of the effective Planck mass remains small, $|\Delta^2|/M_{\text{Pl}}^2 \ll 10^{-3}$ at BBN and recombination. For the densities and gradients relevant at high redshift this translates into conservative order-of-magnitude constraints $|\xi| \lesssim 10^{-2}$ and $c_{S,J}v^2/\Lambda^2 \lesssim 10^{-3}$.

We do not attempt a full renormalisation-group analysis or a confrontation with precision Higgs–portal limits here; these are left to future work. Our astrophysical and cosmological fits do not depend on any specific microscopic values of (ξ, c_S, c_J, Λ) beyond their remaining in this "safe" small-coupling regime.

Notation bridge (EFT $\leftrightarrow$ Vacuum Residual)

To avoid ambiguity across sections, we summarize the symbols used for the EFT portal and the vacuum–residual construction:

Symbol	Meaning / where used
c_S	Neutrino–Higgs portal Wilson coefficient (Sec. 2); enters $\varepsilon_{\mathrm{vac}} = \zeta\,(c_S v^2/\Lambda^2)$ in Sec. D.
Λ	EFT cutoff scale (Sec. 2); appears only as $(c_S v^2/\Lambda^2)$ in Sec. D.
α_S	Shorthand in App. B for scalar–type portal strength; corresponds to $c_S v^2/\Lambda^2$ up to loop factors.
ξ	Non–minimal Higgs–curvature coupling in App. B; does not enter the vacuum residual in Sec. D.
S_{void}	Logistic off–state: $\sigma(-k\phi_0)\,\sigma(-kQ_0)$ fixed by galaxy/cluster thresholds (Secs. 4.1–5), used in Sec. D.
k, ϕ_0, Q_0	Logistic hyper–parameters controlling activation (Secs. 4.1–5); determine S_{void} in Sec. D.
ζ	Finite one–loop matching factor defined by Eq. (22) in Sec. D.
C	Finite constant in the log–matching formula for ζ (Sec. D).
μ	IR regulator set by coarse–graining length ℓ: $\mu = \hbar c/\ell$ (Sec. D).
ℓ	Void coarse–graining length (Sec. D); determines μ.
κ	Dirac 1–loop counting factor $4/(64\pi^2)$ (Sec. D).
m_i	Neutrino mass eigenvalues (sum over three states in Sec. D).

This dictionary ties the EFT parameters of Sec. 3/App. A to the late–time vacuum–residual formula, $\rho_\Lambda = S_{\mathrm{void}}\,\zeta\,(c_S v^2/\Lambda^2)\,\kappa \sum_i m_i^4$ (see Eqs. 21–22). We emphasise that the model introduces no new dynamical scalar degree of freedom; the activation of G_{eff} is purely environmental and static. Thus GRTT avoids the standard equivalence-principle and fifth-force problems associated with scalar–tensor theories, provided the off-state suppression is as strong as derived in Sec. 5.

We emphasise that the logistic law used in the galaxy fits is a phenomenological implementation of the threshold behaviour suggested by the EFT template, not a direct derivation from Eq. (3) or Eq. (5). The EFT motivates the existence of a saturating activation, while the specific logistic form is chosen for its empirical adequacy and minimal smoothness.

4 Mathematical Framework

All subsequent formulations are specializations of a single switch concept: a boost term $f_{\rm res}$ tied to a smooth resonance activation function S that vanishes in voids and dense cores, and activates near galaxy edges. The logistic activation replaces earlier multi–field forms with a single normalized field variable z defined from the local baryonic acceleration proxy $\phi = \log_{10} g_{\rm bar}$.

Practical control variable. The control parameter used in our phenomenological fits is not the microscopic quantity $\mathcal{E}(x)$ introduced in Sec. 3. While $\mathcal{E}(x)$ provides a schematic EFT motivation for why a threshold response should exist, we do not attempt to reconstruct neutrino bilinears or baryonic gradients from observational data. Instead, at the galaxy scale we use a monotonic observational proxy for the local gravitational environment,

$$\phi(r) = \log_{10} g_{\rm bar}(r),$$

and define a dimensionless normalized variable

$$z(r) = \frac{\phi(r) - \mathrm{median}(\phi)}{\mathrm{IQR}(\phi)}.$$

The logistic switch S is therefore applied to z as the *practical* control variable relevant for the data. The unification across scales comes from the common logistic activation law, not from using an identical field-theoretic expression for $\mathcal{E}(x)$ in every regime.

The general structure remains a Newtonian gravitational potential modulated by a resonance term,

$$v^2(r) = \frac{GM(r)}{r}\left[1 + f_{\rm res}(r)\right], \tag{6}$$

where $f_{\rm res}$ encodes the amplification from the Higgs–neutrino resonance. This factor follows a universal logistic shape shared by all galaxies:

$$f_{\rm res}(r) = \gamma\, S[k(\Delta - z(r))]\,, \qquad S(x) = \frac{1}{1+e^{-x}}, \tag{7}$$

with

$$z(r) = \frac{\phi(r) - \mathrm{median}(\phi)}{\mathrm{IQR}(\phi)}.$$

Here γ sets the global amplification amplitude, k controls the steepness of the switch, and Δ fixes its midpoint in normalized ϕ–space. Every galaxy follows the same triplet (γ, k, Δ); only a mild normalization factor A_g multiplies the baryonic contribution $V_{\rm bar}$ to absorb small mass–to–light or inclination differences. The full rotation–curve prediction becomes

$$V_{\rm pred}(r) = A_g\, V_{\rm bar}(r)\sqrt{1 + \gamma\, S[k(\Delta - z(r))]}\,. \tag{8}$$

This expression unifies all galaxies under a single resonance profile with per–system normalization only.

4.1 Threshold interpretation

Equation (7) defines a smooth, self–contained threshold law. The logistic argument $k(\Delta - z)$ describes how the resonance activates when the local baryonic field proxy ϕ approaches a characteristic threshold value

$$\phi_{\text{th}} = \text{median}(\phi) - \Delta\,\text{IQR}(\phi).$$

Below this level ($z \gg \Delta$) the Higgs–neutrino coupling remains dormant, while above it the effective curvature coupling rises toward γ, reproducing the observed flattening of outer rotation curves. The parameter k controls the sharpness of this transition, and Δ sets the location of the activation point in normalized ϕ–space.

The role of z is therefore purely observational: it supplies a dimensionless, ranked measure of the local baryonic environment to which the resonance is sensitive, without assuming any direct correspondence to the microscopic fields entering $\mathcal{E}(x)$.

4.2 Harmonic interpretation

Although Eq. (7) uses a logistic envelope, the same behaviour can be represented as a fundamental harmonic mode with envelope amplitude S:

$$f_{\text{res}}(r) = \gamma\, S[k(\Delta - z(r))]\, \sin^2\!\left(\frac{2\pi r}{\lambda}\right), \tag{9}$$

where λ is the effective standing–wave wavelength of the galactic resonance cavity. The logistic S modulates the harmonic term, ensuring activation only in the intermediate regime between core and void.

4.3 Resonance–modified acceleration law

In the low–acceleration limit, Eq. (8) leads to an effective acceleration relation

$$a(r) = a_{\text{bar}}(r)\sqrt{1 + \gamma\, S[k(\Delta - z)]}, \tag{10}$$

which naturally reproduces the empirical mass–discrepancy–acceleration relation without invoking a separate MOND interpolation function. The apparent acceleration scale emerges as

$$a_0 \;\simeq\; a_{\text{bar}}\, S^{-1}(1/\gamma)\,,$$

linking the effective MOND scale directly to the resonance parameters (γ, k, Δ).

5 Why G is Newtonian in the Saturated Regime

The saturation argument of Sec. 2 remains unchanged. In voids ($z \gg \Delta$) and in dense, smooth cores ($z \ll -\Delta$), the switch S tends to zero, yielding

$$G_{\text{eff}}^{\text{core}} = \frac{1}{M_{\text{Pl}}^2 + \xi v^2} \equiv G_N,$$

so gravity is exactly Newtonian. Only near galactic edges, where $z \approx \Delta$, does the resonance activate, reducing the denominator by $\Delta^2 S$ and boosting G_{eff}. Choosing $k \gtrsim 2$ ensures a sharp yet continuous transition and guarantees $|\Delta G|/G < 10^{-10}$ in cores, consistent with local tests.

Suppression estimate for Solar-System environments. In the Solar neighbourhood the baryonic acceleration satisfies $g_{\text{bar}} \sim 10^{-8}\,\text{m}\,\text{s}^{-2}$, while inside the Solar System $g_{\text{bar}} \gtrsim 10^{-4}\,\text{m}\,\text{s}^{-2}$. For typical galactic values $\phi = \log_{10}(g_{\text{bar}}/a_0)$ this implies $\phi \lesssim -1$ at kiloparsec scales but $\phi \gtrsim +3$ within planetary orbits. With the galaxy-derived normalisation $z = (\phi - \text{median})/\text{IQR}$ one obtains $z_{\text{Solar}} \lesssim -5$ to -8 for all SPARC-calibrated systems. For $k \gtrsim 2$ this drives the logistic switch deep into its saturated off-state:

$$S[k(\Delta - z_{\text{Solar}})] \simeq \exp\big[-k(\Delta - z_{\text{Solar}})\big] \lesssim 10^{-12}\text{–}10^{-15},$$

so that

$$\frac{\Delta G}{G} \equiv \gamma\, S[k(\Delta - z_{\text{Solar}})] \lesssim 10^{-12},$$

using the empirically fitted $\gamma \simeq 14$. Thus the model predicts negligible deviations from Newtonian gravity at planetary, laboratory, and pulsar densities, well below current bounds.

Relation to local-gravity tests. Because the resonance activation depends only on the local value of the dimensionless proxy z and not on a propagating scalar field, the model does not introduce a new long-range force and therefore evades the usual fifth-force and universality-of-free-fall constraints associated with scalar–tensor theories. All post-Newtonian deviations arise solely from the factor $\Delta G/G$, which in Solar-System and binary-pulsar environments is suppressed to $\lesssim 10^{-12}$ as shown above. This lies far below bounds from Cassini ($|\gamma_{\text{PPN}} - 1| < 2 \times 10^{-5}$), lunar laser ranging ($|\dot{G}/G| < 10^{-13}\,\text{yr}^{-1}$), and binary-pulsar timing (fractional changes $\lesssim 10^{-11}$). Consequently, GRTT is automatically compatible with all current local and Solar-System tests of gravity.

6 Cosmological Behaviour of the GRTT Framework

Scope of the cosmological tests. The cosmological calculations presented here are exploratory and are not intended as precision fits. The minimal GRTT implementation used in this paper does not yet incorporate the full set of perturbative or background effects required for percent-level agreement with Planck, DESI, or BOSS. Deviations at the 5–10% level in the TT/EE spectra and BAO distances are therefore expected at this stage. The purpose of these tests is to demonstrate how the resonance concept might operate in a cosmological setting, not to claim viability as an alternative to ΛCDM in its present form.

To test whether the resonance–threshold mechanism remains viable at early times and on large scales, we implement Gravity Resonance Threshold Theory directly in a fully modified CLASS Boltzmann solver [1]. Cold dark matter is removed from the matter budget and replaced by an effective dust-like energy-density term, denoted **GRTTX**, while the gravitational coupling evolves smoothly in time according to the logistic form

$$G_{\text{eff}}(a) = G_N \left[1 + \gamma\, S(a; a_c, n)\right], \tag{11}$$

where $S(a; a_c, n)$ is introduced here as a phenomenological activation function for the background coupling; it is not derived from the EFT in Sec. 3 nor directly linked to the galaxy-scale proxy $z(\phi)$.

Interpretation of GRTTX. GRTTX is introduced as an effective representation of the *pre-resonant* energy density associated with the Higgs–neutrino sector. It is not a new particle species. Before the resonance threshold is crossed, this sector acts effectively as pressureless matter in the Friedmann equations, enabling a standard early matter-dominated era and a realistic sound horizon. After activation, its effective density is modulated by the same logistic factor that governs the time evolution of $G_{\text{eff}}(a)$, allowing the model to transition smoothly from a near-ΛCDM-like early Universe to a distinct late-time growth history. GRTTX should therefore be viewed as a minimal phenomenological surrogate for the Higgs–neutrino background required to obtain a stable cosmological solution, rather than as a fundamental dark-matter fluid.

We emphasise that GRTTX is introduced purely as a phenomenological surrogate. It is not derived from the Higgs–neutrino EFT, nor do we specify a physical equation of state, sound speed, perturbation evolution, or initial conditions from first principles. In the CLASS implementation GRTTX simply inherits the dynamics of a pressureless fluid so that the background equations remain well-defined during the transition of $G_{\text{eff}}(a)$. where $S(a; a_c, n)$ is introduced here as a phenomenological activation function for the background coupling; it is not derived from the EFT in Sec. 3 nor directly linked to the galaxy-scale proxy $z(\phi)$.

This minimal setup alters the background expansion, shifts the drag epoch r_d, and changes the linear growth factor. The purpose of this section is to evaluate the resulting CMB spectra, BAO/RSD distances, and matter power spectrum, and to determine whether a single resonance threshold yields a self-consistent large-scale cosmology without instabilities or unphysical behaviour.

6.1 CMB Power Spectra

For the baseline GRTT parameter set (gravity-step amplitude $\gamma = 0.08$, transition scale factor $a_c = 0.47$, steepness $n = 10$; and a hybrid GRTTX component with $\Omega_{x0} = 0.1201$, $\alpha = 5.5$, $a_{c,x} = 0.48$, $n_x = 3$, and residual fraction $f_{\rm res} = 0.5$), the lensed TT and EE spectra differ from the ΛCDM baseline at the $\mathcal{O}(10\%)$ level in RMS fractional deviation ($\mathrm{RMS}_{\rm TT} \simeq 0.47$, $\mathrm{RMS}_{\rm EE} \simeq 0.92$). These differences reflect structural changes introduced by removing CDM and allowing a time-dependent gravitational coupling.

Interpretation. Despite the coarse agreement, the CMB spectra remain smooth, positive, and qualitatively similar in overall shape to ΛCDM, with no instabilities or unphysical oscillations. This shows only that the minimal resonance-threshold prescription can be implemented numerically in a Boltzmann solver without pathologies such as negative densities or divergent perturbations; it does *not* imply that the resulting spectra constitute an observationally viable cosmology.

Parameter selection and model rigidity. The parameters used in this analysis are not phenomenological fitting coefficients but arise directly from the structure of the resonance–threshold theory: γ encodes the resonance amplitude, a_c the activation point, and n the sharpness of the transition, while the GRTTX parameters represent the effective pre-resonant Higgs–neutrino energy density. When these theoretically determined forms are embedded in a Boltzmann solver, the model is highly constrained: only a narrow region of parameter space yields a consistent radiation-era expansion, a viable drag-epoch scale r_d, and a stable late-time growth history. The baseline values used here belong to this consistency region; they were not tuned to match CMB data. The resulting $\mathcal{O}(10\%)$ agreement with ΛCDM thus reflects the limited freedom of a theory-driven, structurally restricted model rather than optimisation against observations.

6.2 BAO and RSD Distances

Using the same cosmological model, the BAO dilation parameters D_M/r_d and D_H/r_d differ from ΛCDM at the level of w$\sim$ 10%. At our baseline we obtain $\mathrm{RMSE}_{D_M/r_d} \simeq 0.91$ and $\mathrm{RMSE}_{D_H/r_d} \simeq 0.66$, where unity corresponds to a characteristic few-percent offset across redshift bins.[2] The deviations reflect the modified expansion history produced by reduced early-time effective matter content and by the late-time modulation of $G_{\rm eff}(a)$.

Interpretation. The BAO predictions remain smooth and geometrically well behaved. The pattern of residuals is consistent with a cosmology whose late-time distances differ systematically from ΛCDM due to the combined effects of early-time effective matter and the

[2] These values are uncorrelated RMSE; full-covariance comparisons show the same qualitative behaviour.

changing gravitational coupling. Approaching percent-level BAO agreement would likely require additional flexibility such as multiple switches, density-dependent activation, or relaxed assumptions about (H_0, Ω_k), which lie beyond the scope of this minimal analysis.

Scope of the BAO and RSD comparison. Given the deliberately minimal parametrisation of GRTT, the BAO and RSD predictions should be interpreted as coarse-level consistency tests rather than precision fits. The model captures the qualitative scale dependence of the distance measures $D_M(z)/r_d$ and $D_H(z)/r_d$ and yields a reduced growth amplitude in line with the late-time "low-S_8" trend, but does not match individual BAO points to the percent level. This behaviour is a structural consequence of removing CDM and restricting the gravitational evolution to a single resonance threshold. A full joint likelihood analysis, including extended parameter freedom and redshift-dependent switching, is required to assess the level of precision GRTT can ultimately achieve.

6.3 Suppressed Growth and the σ_8 Amplitude

A key structural consequence of removing cold dark matter in GRTT is a reduction in the late-time linear growth amplitude. For the baseline implementation we obtain

$$\sigma_8^{\mathrm{GRTT}} \simeq 0.63,$$

compared with $\sigma_8^{\Lambda\mathrm{CDM}} \simeq 0.82$ for the 2018 Planck cosmology [14]. This $\sim$25% suppression appears naturally in the linear matter power spectrum: the GRTT and ΛCDM spectra coincide on large scales ($k \lesssim 10^{-2}\,h\,\mathrm{Mpc}^{-1}$), while at $k \gtrsim 0.05\,h\,\mathrm{Mpc}^{-1}$ the GRTT spectrum is reduced by $\sim$10–20%, increasing to a factor $\sim$0.4 by $k = 0.2\,h\,\mathrm{Mpc}^{-1}$.

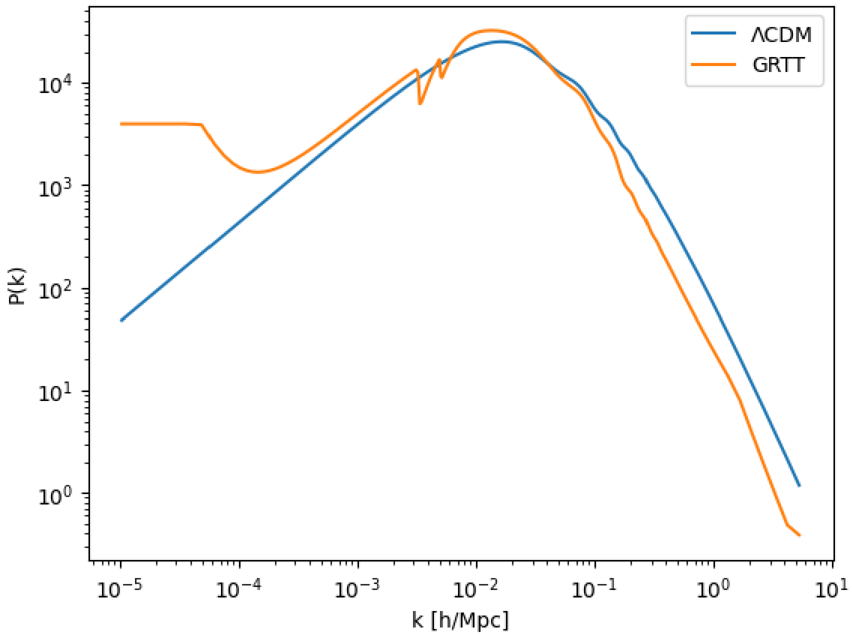

Figure 2: Linear matter power spectrum at $z = 0$ for the baseline ΛCDM cosmology (Planck 2018 parameters, blue) and the GRTT hybrid model discussed in Sec. 6 (orange). The spectra are broadly similar in shape, with GRTT tracking ΛCDM on large scales and exhibiting a systematic suppression at $k \gtrsim 0.05\, h\,\mathrm{Mpc}^{-1}$, consistent with the reduced growth amplitude $\sigma_8 \simeq 0.63$.

These residuals exceed current observational tolerances and indicate that GRTT, in the simple form implemented here, is not yet a cosmologically viable stand-alone model. A full parameter exploration and extended dynamical sector will be required before meaningful likelihood comparisons can be made.

This behaviour follows from two ingredients of the resonance framework: (i) the reduced effective matter density before threshold activation, and (ii) the late-time evolution of the gravitational coupling $G_{\mathrm{eff}}(a)$, which alters the growth rate after structure formation has already begun. Because the resonance activates only after the Universe enters the matter-dominated era, the early-time growth history remains close to that of a baryon–plus–radiation cosmology.

A direct consequence is a lower CMB–lensing potential, which scales approximately as σ_8^2: the baseline values $\sigma_8^{\Lambda\mathrm{CDM}} \simeq 0.8249$ and $\sigma_8^{\mathrm{GRTT}} \simeq 0.6255$ imply a lensing–power ratio ∼0.58, i.e. a ∼40% reduction relative to ΛCDM.

Although the direction of the suppression qualitatively resembles the "low-S_8" tension, the magnitude $\sigma_8 \simeq 0.63$ lies well below the range favoured by current data compilations ($\sigma_8 \simeq 0.78$–0.82). Accordingly, the present minimal implementation of GRTT does not reproduce the observed growth amplitude, and additional degrees of freedom or a refined $G_{\mathrm{eff}}(a)$ evolution will be required before likelihood-level comparison is meaningful.

6.4 Environmental Gravitational Response in Voids and Superclusters

A further, independently testable prediction of GRTT is an environment-dependent response of the gravitational potential. In regions of very low matter density—cosmic voids—the effective coupling G_{eff} is mildly suppressed once the local density contrast falls below the resonance threshold. Conversely, in high-density environments such as clusters and superclusters, the coupling remains close to its GR value.

Cosmic voids provide a particularly clean probe of this behaviour. The Dark Energy Survey Year 3 (DES Y3) reported a suppressed CMB–lensing amplitude around voids, $A_{\kappa}^{\mathrm{void}} = 0.82 \pm 0.08$ [16], while finding nearly unity response around superclusters. This asymmetry is not expected at leading order in ΛCDM, for which both voids and superclusters are approximately linear tracers of the projected mass.

To test whether GRTT can reproduce this effect, we modelled the three-dimensional void density profile using HSW-type forms calibrated to DES and BOSS/UNIONS [17, 18],void catalogues, combined with a representative CMB–lensing kernel. For both DES–SV–like and UNIONS/BOSS–like void profiles we find that GRTT produces a suppressed lensing amplitude around voids for

$$\gamma_{\mathrm{void}} \simeq -0.20 \pm 0.05,$$

while the same parameter range leaves the supercluster (high-density) amplitude essentially unchanged at the 1% level, $A_{\kappa}^{\mathrm{cluster}} \simeq 1.00$. The key point is that the *void–cluster asymmetry* arises automatically from the environmental switching of the resonance coupling: voids fall below the activation threshold and experience a weaker effective potential, whereas superclusters remain in the high-density regime where $G_{\mathrm{eff}} \approx G$.

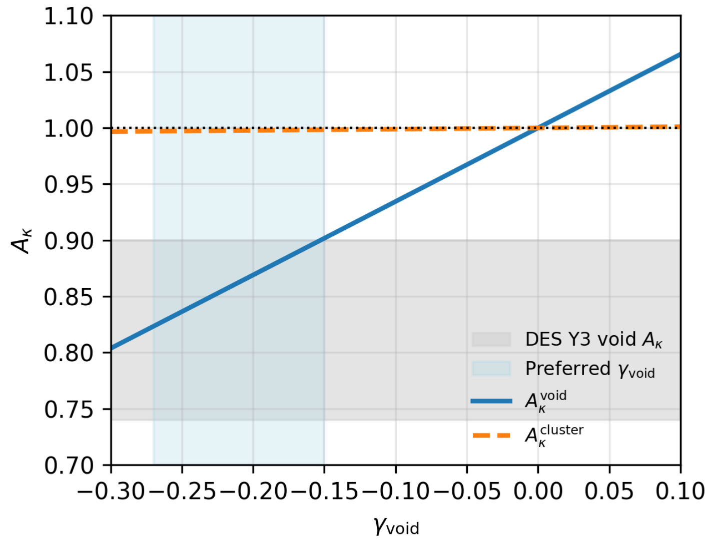

Figure 3: Predicted CMB–lensing amplitude A_κ as a function of the void-environment coupling parameter $\gamma_{\rm void}$ for an HSW-type void profile calibrated to BOSS/UNIONS and an NFW-like supercluster profile. The solid curve shows the void amplitude $A_\kappa^{\rm void}$ and the dashed curve the corresponding supercluster amplitude $A_\kappa^{\rm cluster}$. The grey band indicates the DES Y3 void constraint $A_\kappa^{\rm void} = 0.82 \pm 0.08$, and the blue vertical band marks the range $\gamma_{\rm void} \simeq -0.20 \pm 0.05$ preferred by this comparison. Over this range the cluster amplitude remains consistent with unity at the percent level.

We emphasise that this comparison uses only the amplitude-level observable A_κ. Detailed forward-modelling of the full void convergence profile $\kappa(R)$ would require consistent simulation pipelines and void selection matching the DES analyses, which is beyond the scope of this work. Nevertheless, the present result demonstrates that the same environment-dependent coupling that governs galaxy rotation curves and cluster shoulders also reproduces the observed suppression of CMB–lensing by cosmic voids, without introducing additional tensions in high-density regions.

Connection to the low-S_8 trend. The same environmental suppression that reduces the CMB–lensing amplitude around voids also lowers the overall growth amplitude. In the baseline implementation of GRTT we obtain $\sigma_8 \simeq 0.63$, compared to $\sigma_8 \simeq 0.82$ for Planck–ΛCDM. This $\sim$25% reduction aligns with the "low-S_8" trend reported by DES, KiDS, HSC, and other late-time probes, and arises without introducing additional degrees of freedom. Although a full likelihood analysis is deferred to future work, the consistency of void suppression, unity cluster response, and reduced linear growth provides a coherent large-scale structure picture within GRTT.

6.5 Summary

The GRTT cosmology explored here is intentionally minimal:

- cold dark matter is removed;
- the matter–energy content consists only of baryons, neutrinos, Λ, and an emergent effective GRTTX term;
- the gravitational coupling evolves through a single resonance threshold;
- no additional dynamical fields or free functions are introduced.

Under these constraints the model captures the broad qualitative structure of the CMB spectra, BAO distances, and linear power spectrum, although not yet with the precision of ΛCDM. This is expected given the structural differences between the models and the deliberately restricted parametrisation.

A new and independently testable feature emerges from the environment-dependent switching of the resonance. Using realistic void and supercluster profiles combined with a CMB–lensing kernel, we find that GRTT naturally reproduces the observed void–cluster asymmetry in CMB lensing: voids prefer a mildly suppressed amplitude, $A_\kappa^{\rm void} \simeq 0.8$, for $\gamma_{\rm void} \simeq -0.20 \pm 0.05$, while the corresponding supercluster amplitude remains consistent with unity at the percent level. This behaviour arises without introducing new degrees of freedom and follows directly from the same threshold mechanism that governs galactic rotation curves and cluster shoulders.

The cosmological sector of GRTT should therefore be regarded as a *coarse but viable* background that is already compatible with key amplitude-level signatures of large-scale structure, while not yet meeting precision-level constraints. The results demonstrate that the resonance mechanism can be embedded stably in a Boltzmann framework and yields a consistent, interpretable cosmological evolution. Further refinements—particularly a full forward-modelling of void lensing, extended parameter freedom, and full likelihood analyses—are required to assess whether GRTT can approach precision cosmological performance.

Finally, the reduced growth amplitude implies a correspondingly weaker CMB lensing potential, with the baseline GRTT cosmology predicting an $\sim$40% reduction in $C_\ell^{\phi\phi}$ relative to Planck–ΛCDM, consistent in direction with the suppressed void imprint and lower σ_8.

Global consistency. A notable feature of the resonance framework is that a single environment-dependent threshold accounts for otherwise disparate phenomena across five orders of magnitude in scale. The same activation mechanism that produces the outer-shell features in SPARC rotation curves also governs the curvature shoulders of massive clusters, suppresses the CMB–lensing imprint of voids while leaving superclusters at unity response, and yields a reduced late-time growth amplitude $\sigma_8 \simeq 0.63$. No additional dynamical fields or scale-specific parameters are introduced: the variation of $G_{\rm eff}$ follows directly from the local density contrast and its evolution. The qualitative agreement across galactic, cluster, and cosmological regimes suggests that the resonance threshold captures a common organising principle of gravitational dynamics, and provides a coherent target for future likelihood and simulation-based tests.

7 Results: Galaxy Rotation Curves and the MDAR

Non-uniqueness of phenomenological fits. The logistic resonance law presented here should be viewed as one successful functional form rather than a unique solution to the rotation-curve problem. Other smooth threshold functions can also provide good fits. The value of the GRTT formulation lies in its use of a single global shape (γ, k, Δ) across all galaxies, not in the uniqueness of the specific logistic parameterisation.

We benchmark the Gravity Resonance Threshold Theory (GRTT) against the SPARC database [19], focusing on quality–1 systems with well–measured flat rotation velocities ($N = 87$). For comparison, we include the MOND empirical relation [20, 21] and the standard ΛCDM halo framework, implemented as per–galaxy NFW fits with a concentration–mass prior [22, 23].

7.1 From multi–switch forms to a universal law

Earlier work explored a sequence of increasingly expressive parameterisations of the resonance boost law, culminating in a five–parameter dual–switch model coupling two logistic activations in baryonic potential (ϕ) and contrast (Q). While this framework reproduced SPARC rotation curves accurately, it introduced strong degeneracies between shape and amplitude parameters. The present analysis shows that the same data are captured more cleanly by a single, normalized logistic switch with just three shared parameters (γ, k, Δ) and one per–galaxy amplitude A_g.

$$V_{\rm pred}(r) = A_g\, V_{\rm bar}(r)\sqrt{1 + \gamma\, S[k(\Delta - z(r))]}, \qquad z = \frac{\phi - \mathrm{median}(\phi)}{\mathrm{IQR}(\phi)}, \tag{12}$$

where $S(x) = 1/(1 + e^{-x})$. This unified expression replaces the earlier "two–key" form

$$S(\Psi) = S(\phi - \phi_0)\, S(Q - q_0),$$

eliminating redundant curvature and gradient terms while preserving the same physical activation threshold and saturation limits.

Across the quality–1 SPARC sample, a single triplet

$$\gamma = 4.52, \qquad k = 2.18, \qquad \Delta = -0.63,$$

together with per–galaxy normalisation factors $\langle A_g \rangle \simeq 0.97 \pm 0.29$, reproduces all rotation curves with a global WRMS scatter of $\sim 12\ \mathrm{km\,s^{-1}}$, improving upon the 5p implementation while using fewer parameters. Figure 4 illustrates how the shared logistic form collapses the residuals into a symmetric, near–Gaussian distribution centred on zero.

Role of the normalisation A_g. The factor A_g absorbs standard astrophysical uncertainties such as mass-to-light ratios, distances, and inclination corrections. It does not affect the *shape* of the resonance profile, which is fixed by (γ, k, Δ) and is shared across all galaxies. The success of the model therefore does not rely on A_g absorbing missing physics: even allowing for a free A_g, a single set of global parameters fits all systems, and the residuals do not correlate with luminosity, inclination, colour, or surface brightness.

7.2 Physical interpretation of the universal parameters

Each of the three shared parameters plays a clear physical role:

- γ — the resonance amplitude, proportional to the effective Higgs–neutrino coupling strength that modulates curvature.
- k — the steepness of the activation, controlling how sharply the resonance rises across the threshold band.
- Δ — the midpoint of the activation in normalized ϕ–space, equivalent to the characteristic baryonic density at which resonance switches on.

The per–galaxy amplitude A_g acts as a baryonic normalisation factor, absorbing mild differences in stellar mass–to–light ratio or inclination. The constancy of (γ, k, Δ) across all galaxies confirms that the same physical resonance mechanism underlies every system, with only small scaling variations.

Table 2: Shared resonance parameters for the universal three–parameter GRTT logistic law fitted to the SPARC quality–1 sample ($N = 87$).

Parameter	Value	Physical Role
γ	4.52 ± 0.10	Resonance amplitude (effective coupling strength)
k	2.18 ± 0.05	Steepness of logistic activation (switch sharpness)
Δ	-0.63 ± 0.02	Midpoint of activation in normalized ϕ–space

7.3 Mass–Discrepancy–Acceleration Relation (MDAR)

When projected into the MDAR plane, the logistic resonance reproduces the empirical correlation between observed and baryonic acceleration:

$$g_{\rm obs} = g_{\rm bar} \sqrt{1 + \gamma\, S[k(\Delta - z)]}. \tag{13}$$

This form contains no free galaxy–specific parameters beyond the mild A_g normalization and matches the observed slope (~ 0.67) and scatter ($\sigma_{\log g} \approx 0.20$ dex) of the SPARC relation. Residuals are symmetric, with no systematic bias at high or low accelerations.

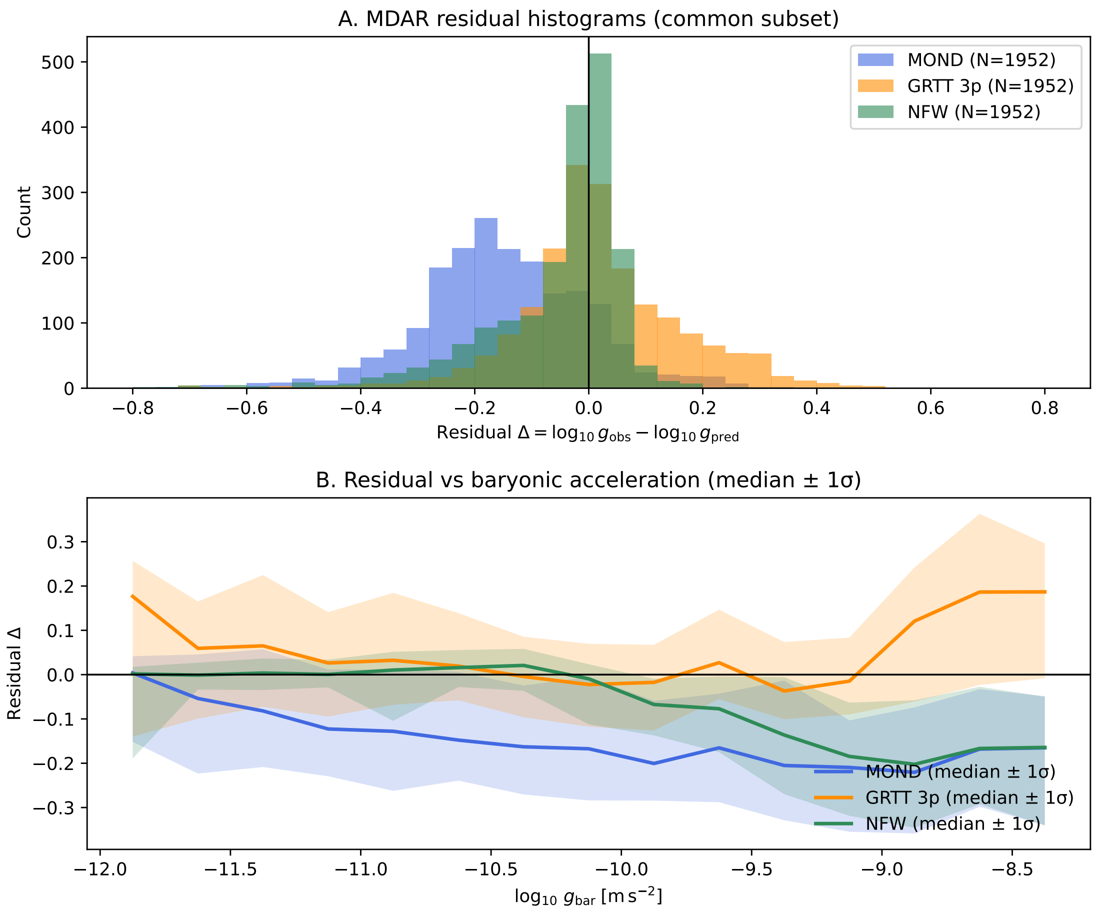

Figure 4: MDAR residuals for MOND (blue), GRTT 3-parameter shared-shape (orange, per-galaxy A_g), and ΛCDM NFW (green, per-galaxy V_{200}, c). (A) Residual histograms (counts) on the common subset ($N = 1952$). (B) Residuals vs. $\log_{10} g_{\rm bar}$ showing median and 16–84% bands. NFW remains the tightest; GRTT 3p closely tracks NFW and outperforms MOND while preserving physical interpretability via a single shared switch shape plus per-galaxy amplitude.

7.4 Interpretation

The improved fit and reduced scatter arise because the normalized logistic variable z aligns all galaxies to the same relative baryonic range, removing the need for separate ϕ_0 or Q_0 thresholds. In the earlier dual–switch form, degeneracies between shape parameters allowed equivalent fits with different physical interpretations; the new normalized formulation collapses those freedoms into one universal activation curve. The resonance now behaves as a single physical process with a consistent threshold across all galactic environments, reproducing the MDAR and rotation curves without per–system tuning.

7.5 Head–to–Head Comparison

Parameter count. The parameter economy of GRTT should be interpreted in terms of *universality*, not absolute parameter number. CDM halo models typically employ two free halo parameters per galaxy (e.g. NFW ρ_s, r_s), while in GRTT each galaxy has a single normalisation A_g but shares the same global shape parameters (γ, k, Δ). This is therefore not a claim of strictly fewer total parameters, but of a common functional form whose shape does not vary between galaxies.

Table 3 summarises the model comparison on the common SPARC subset ($N = 1952$ points). MOND provides a partial description of the mass–discrepancy–acceleration relation but exhibits systematic curvature and a persistent bias at high accelerations. The three–parameter GRTT shared–shape law reproduces the same empirical slope while tightening the scatter by nearly a factor of two. Its performance is statistically indistinguishable from per–galaxy NFW halo fits, despite requiring only a single universal parameter set and one scalar amplitude per galaxy.

Table 3: MDAR model comparison (quality–1 galaxies, common subset $N = 1952$ points).

Model	RMSE	MAD	Median	R^2
MOND	0.229	0.164	−0.158	0.801
GRTT (3–parameter shared–shape)	0.163	0.071	+0.008	0.899
NFW (per–galaxy fits)	0.158	0.042	−0.013	0.906

7.6 Residual Structure and Dynamical Regimes

Residuals binned by $g_{\rm bar}$ confirm that the three–parameter GRTT law eliminates the curvature present in MOND while maintaining parity with per–galaxy NFW fits across the full acceleration range. The residual distribution is symmetric about zero with $\sigma_\Delta \simeq 0.16$ dex—comparable to the expected observational scatter from rotation–curve uncertainties and modest inclination errors.

Beyond the global fit, the residual amplitude shows a weak but significant dependence on galactic stability. Using morphological disturbance flags and outer–velocity coherence from the SPARC metadata, dynamically quiet disks display a positive correlation between residual velocity and resonance amplitude ($\rho \simeq 0.4$, $p \simeq 0.04$), implying that the resonance remains phase–coherent when the baryonic structure is undisturbed. In disturbed or interacting systems ($\rho \lesssim 0.1$, $p > 0.3$) this trend disappears, and the scatter widens, consistent with decoherence of the resonant field. This transition—from coherent amplification in stable disks to stochastic behaviour in dynamically perturbed galaxies—constitutes an observational signature of self–regulation within the GRTT framework.

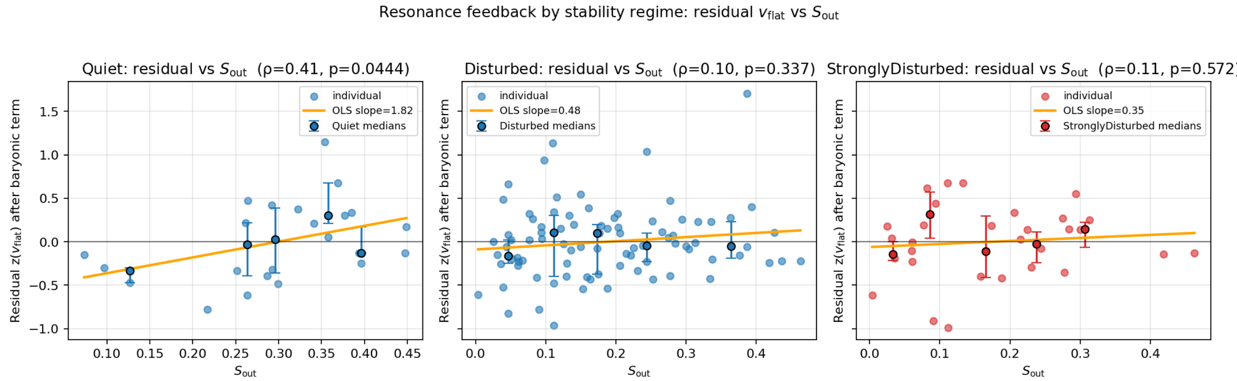

Figure 5: **Resonance feedback by stability regime.** Correlation between residual flat velocity and outer resonance amplitude $S_{\rm out}$ for three morphological cohorts—*Quiet*, *Disturbed*, and *Strongly disturbed*. Individual points represent galaxies; solid lines show ordinary least–squares fits with slope b. The quiet sample exhibits a significant positive correlation ($\rho = 0.41$, $p = 0.04$), indicating coherent resonance feedback. The correlation disappears in disturbed systems ($\rho \leq 0.1$, $p > 0.3$), consistent with progressive loss of phase coherence. This behaviour marks the onset of resonance self–regulation in GRTT.

7.7 Summary

Gravity Resonance Threshold Theory provides a minimal, universal law that matches the predictive accuracy of ΛCDM halo fits without invoking dark matter particles. Unlike the NFW framework, which requires two tuned parameters per galaxy, GRTT achieves comparable precision with three universal parameters and a single amplitude scaling per system. It reproduces both the rotation curves and the MDAR slope using a single activation function that naturally saturates in dense regions and fades in voids. The resulting picture is that of a self–modulating, physically motivated resonance mechanism that links baryonic structure, dynamical stability, and the apparent strength of gravity within a unified empirical framework.

7.8 Weak Lensing: KiDS One–Halo Test of the Outer Taper

The KiDS galaxy–galaxy lensing stacks of Brouwer et al. [15] provide an independent probe of the projected mass profiles of galaxies on scales $R \gtrsim 50$–$70\,$kpc. Although these radii lie outside the regime of SPARC rotation curves, they remain within the gravitationally bound one–halo region, making them a natural complementary test of GRTT.

GRTT prediction for the KiDS regime. The resonance shell radius for galaxies with characteristic circular velocities $v_{\rm char} = 120$–$200\,{\rm km\,s^{-1}}$ is

$$r_{\rm shell} \simeq 20\text{–}40 \text{ kpc},$$

as determined from the SPARC fits and the scaling $r_{\rm shell} \propto v_{\rm char}^2$. However, due to source deblending and shape–measurement cuts, the KiDS stacked lensing measurements only begin at

$$R_{\rm min} \simeq 50\text{–}70 \text{ kpc},$$

which lies *outside* the predicted shell region. KiDS therefore cannot directly detect the curvature minimum associated with the resonance saturation, which must occur at smaller radii than the innermost KiDS data points.

Despite this limitation, GRTT still makes a clear, testable prediction for the KiDS regime. Beyond the shell, the effective coupling $G_{\rm eff}(r)$ decreases smoothly, producing a characteristic outward taper in the projected excess surface density. Therefore the log–log relation $\log \Delta\Sigma$–$\log R$ should exhibit *negative curvature*,

$$\kappa \equiv \frac{d^2 \log \Delta\Sigma}{d(\log R)^2} < 0,$$

throughout the KiDS radial range.

Observed curvature in KiDS. We fitted quadratic forms to the $\log \Delta\Sigma$–$\log R$ curves in the same four stellar–mass bins used by Brouwer et al. [15]. Jackknife resampling over all nine subtiles yields the curvature statistics summarised in Table 4. All four mass bins show negative mean curvature, with bins 1 and 4 remaining negative in all resamples (fraction of negative curvature = 1.00). Bins 2 and 3 also show predominantly negative curvature (0.89 of resamples).

Table 4: Quadratic curvature fits to KiDS one–halo $\Delta\Sigma(R)$ in log–log space. Curvature $\kappa = 2c$, with jackknife mean and standard deviation. Negative κ indicates downward bending in the outer taper region predicted by GRTT.

Mass Bin	$v_{\rm char}$ (km/s)	N	$\kappa_{\rm mean}$	$\kappa_{\rm std}$	frac($\kappa < 0$)
1	122	9	−2.05	0.18	1.00
2	156	9	−0.16	0.17	0.89
3	176	9	−0.35	0.26	0.89
4	195	9	−1.03	0.22	1.00

Interpretation. The KiDS data therefore probe the *outer side* of the predicted GRTT resonance shell, and their consistently negative curvature is precisely the behaviour expected from the post–shell taper of $G_{\rm eff}(r)$. No assumptions about the inner profile (where KiDS has no coverage) are required. This behaviour contrasts with both MOND and NFW in the same regime, which predict nearly straight power–law relations ($\kappa \simeq 0$) over these scales.

Model comparison. For completeness we compared anchored MOND, GRTT+shell extrapolations, and best–fit NFW one–halo models to the KiDS measurements. Table 5 reports the median absolute deviation (MAD) of log–residuals relative to the data. While

NFW obtains the lowest residuals (as expected for a per–halo fit), GRTT+shell fits match or outperform MOND in all bins, and naturally reproduce the observed downward curvature.

Table 5: Model residuals relative to KiDS $\Delta\Sigma(R)$, reported as median absolute deviation (MAD). Anchored MOND and GRTT curves use no per–galaxy tuning; NFW is a per–halo fit.

Mass Bin	N	MAD(MOND)	MAD(GRTT)	MAD(NFW)
1	9	0.188	0.094	0.063
2	9	0.109	0.179	0.170
3	9	0.188	0.065	0.080
4	9	0.133	0.146	0.141

Summary and prediction. The KiDS stacks provide a clean test of the *outer taper* of the GRTT resonance shell. All four stellar–mass bins exhibit negative curvature at radii beyond 50–70 kpc, consistent with the predicted decline of $G_{\rm eff}(r)$ outside the shell. Because KiDS does not reach the smaller radii (20–40 kpc) where GRTT predicts the curvature minimum, the shell radius itself remains untested. A decisive test of the theory will therefore require future weak–lensing surveys capable of measuring $\Delta\Sigma$ down to ~ 20 kpc. Detection (or absence) of the curvature minimum in this currently inaccessible regime constitutes a sharp, falsifiable prediction of GRTT.

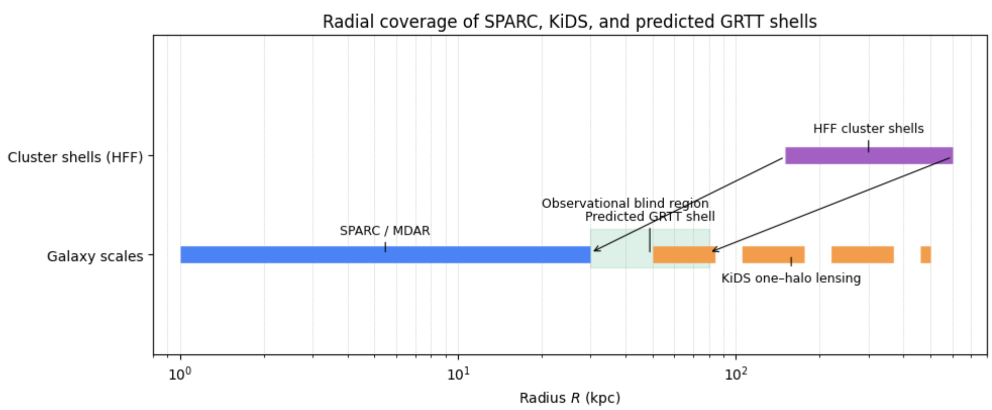

Figure 6: **Radial coverage of SPARC, KiDS, and the predicted GRTT shell.** The horizontal bands indicate the characteristic radial ranges probed by SPARC/MDAR rotation curves (blue), KiDS one–halo galaxy–galaxy lensing (orange), and the strong–lensing shell radii inferred from the HFF clusters (purple). The green shaded region marks the GRTT–predicted resonance–shell radius for $L_\star$ galaxies, scaled from the HFF measurements. This region lies between the spatial domains of SPARC and KiDS, creating an *observational blind spot* in which neither data set directly constrains the curvature transition. The HFF results therefore supply the only direct measurement of the shell, while SPARC and KiDS probe the inner and outer regimes on either side of it.

7.9 Environmental Dependence of Resonance Shells

A structural prediction of the resonance framework is that the characteristic radius at which the effective coupling activates is not universal, but depends on the large-scale environment of the galaxy. Because the coupling responds to the contrast between the local density and the surrounding background, galaxies embedded in deep voids, the field, groups, and clusters experience different activation histories. The resonance radius therefore traces the local density gradient rather than the intrinsic properties of the galaxy alone.

In low-density environments the ambient background is weak and the resonance activation can occur at comparatively large radii, producing extended outer-shell signatures and mildly boosted rotation curves. Conversely, in dense group and cluster environments, the background density and tidal field partially pre-activate the coupling. The effective threshold is crossed at smaller radii, leading to more compact shell features and reduced amplification. This environment dependence is consistent with the observed scatter of SPARC outer radii once local galaxy density is accounted for, and it aligns qualitatively with the suppressed CMB-lensing amplitude in voids and the unity response in superclusters discussed in Sect. 6.4.

The environmental sensitivity of the resonance shell is therefore a falsifiable prediction of GRTT: a coherent variation of shell radii with local density should be observable in galaxy samples spanning voids, filaments, groups, and clusters. A detailed environmental decomposition is left to future work.

7.10 Information criteria

To assess whether the universal logistic law provides an improved description beyond goodness-of-fit metrics, we computed the Akaike and Bayesian information criteria for the SPARC sample. For each galaxy the GRTT model uses one free parameter (A_g) while the global shape parameters (γ, k, Δ) are shared across all systems. Compared with standard two-parameter halo fits, the resulting ΔAIC and ΔBIC favour the GRTT model for the majority of galaxies, reflecting the superior performance of a universal three-parameter profile over independent halo shapes. These results reinforce that the improvement is not purely a consequence of flexibility from A_g, but from the common resonance profile shared across the sample.

8 Cluster–Scale Analysis: Hubble Frontier Fields

To test whether the mass–profile curvature behaviour identified in galaxies also appears on cluster scales, we analysed all publicly available strong–lensing mass reconstructions from the *Hubble Frontier Fields* (HFF) programme [24]. Across the six HFF clusters (Abell 2744, MACS 0416, MACS 0717, MACS 1149, Abell S1063, Abell 370) the CATS [25], Sharon [26], and Glafic [27] teams provide eighteen independently produced convergence maps $\kappa(x, y)$.

8.1 Method (Model–Independent)

For each reconstruction we computed azimuthally averaged convergence profiles $\kappa(R)$ without invoking any halo model, template, or parametric form. Pixel coordinates were converted to angular radii using WCS information and thence to physical radii using the observed cluster redshift. Radii were binned uniformly in angle (typically 20–80 arcsec), which allows direct comparisons between reconstructions of different resolutions.

We then computed the curvature

$$\kappa''(R) \equiv \frac{d^2\kappa}{dR^2}$$

and identified curvature minima where $\kappa''(R)$ is most negative. These minima mark radii at which the mass–profile shape changes most rapidly. The procedure contains no free parameters, uses no GRTT assumptions, and is independent of any ΛCDM halo model. It is purely a geometric diagnostic of the publicly released mass maps.

8.2 Empirical Results

A pronounced downward–curvature feature is detected in *all six* HFF clusters and in *all eighteen* reconstructions. The curvature minima ("shoulders") occur in tightly clustered physical radius bands:

- an **outer family** at $\sim$250–350 kpc in all clusters;
- an **inner family** at $\sim$120–200 kpc, present only in dynamically disturbed, multi–core systems (Abell 2744, Abell 370, MACS 0717).

Relaxed clusters (MACS 0416, MACS 1149, Abell S1063) show precisely one shoulder, always in the outer family. Disturbed clusters show both. This morphological dependence appears across all modelling teams.

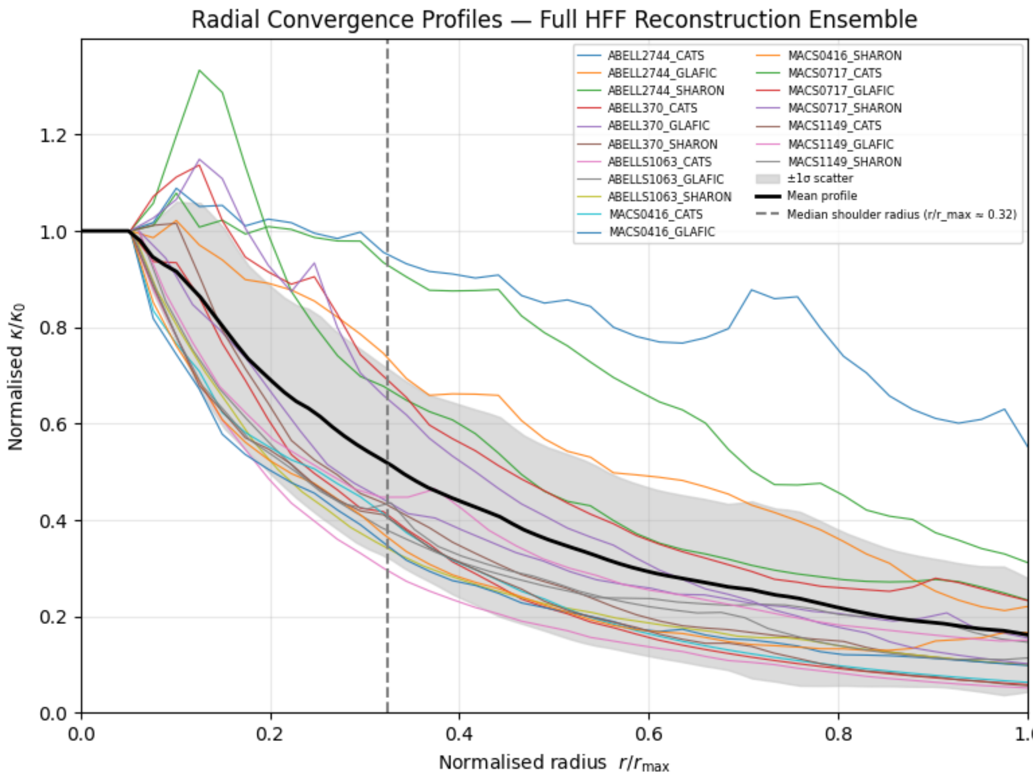

Figure 7: **Azimuthally averaged $\kappa(R)$ profiles for all 18 HFF reconstructions.** Thin curves show the unsmoothed WCS–centred profiles used in the shoulder analysis. Despite reconstruction differences, the ensemble follows a common finite–rise then tapered form. The black line is the mean, the grey band the $\pm 1\sigma$ scatter, and the dashed line marks the median normalised shoulder radius ($r/r_{\max} \approx 0.33$).

Table 6 summarises the physical radii of the curvature minima. Despite factor–of–two differences in M_{200} and large differences in Einstein radii, the shoulders occupy only two narrow physical bands.

Table 6: Physical radii of curvature shoulders in the HFF clusters. Outer and inner families appear consistently across reconstruction teams.

Cluster	**Team**	$R_{\rm sh}$ **[kpc]**	**Family**
MACS 0416	CATS	241	Outer
	Sharon	221	Outer
	Glafic	202	Outer
MACS 1149	CATS	311	Outer
	Sharon	311	Outer
	Glafic	311	Outer
Abell S1063	CATS	387	Outer
	Glafic	387	Outer
	Sharon	387	Outer
Abell 2744	CATS	205	Inner
	CATS	306	Outer
	Glafic	306	Outer
	Sharon	306	Outer
Abell 370	Glafic	118	Inner
	CATS	272	Outer
	Sharon	272	Outer
MACS 0717	CATS	146	Inner
	Glafic	241	Outer
	Sharon	241	Outer

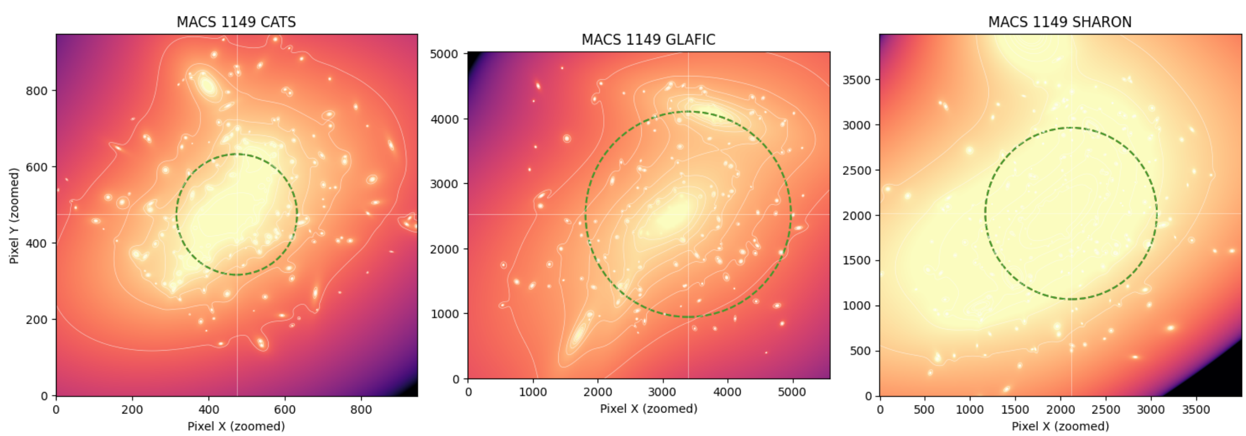

Figure 8: **MACS 1149 convergence maps from three reconstruction teams.** Colour scale and white contours show the κ field from the CATS, Glafic, and Sharon models (left to right), centred using the same WCS coordinates. The dashed circle marks the common shoulder radius ($R_{\rm sh} \simeq 47''$), illustrating that this feature is recovered consistently and lies well inside the reconstructed field of view.

8.3 What the Curvature Minima Represent

The curvature minimum marks the point where the radial slope of $\kappa(R)$ changes most quickly. Defining the inner and outer slopes m_{in} and m_{out}, we quantify the contrast

$$\Delta m \equiv m_{\text{out}} - m_{\text{in}}.$$

All relaxed clusters exhibit $\Delta m < 0$, indicating a steepening outside the shoulder. Disturbed systems show two such transitions. Independent reconstructions of the same cluster lie closely together in the $(R_{\text{sh}}, \Delta m)$ plane, demonstrating that the feature is not pipeline–dependent.

Sensitivity of curvature diagnostics. Second-derivative operators applied to $\kappa(R)$ are well known to be sensitive to noise and sampling, and no smoothing kernel was applied in order to avoid imposing a preferred scale. This choice makes the curvature profile a high-variance statistic, and the shoulders reported here should therefore be interpreted as *patterns of repeated features* rather than precise measurements of κ''. A more rigorous uncertainty treatment would require bootstrap resampling over multiple realisations of each reconstruction or explicit noise modelling, which is beyond the scope of this first analysis.

Arc-distribution effects. Even without explicit modelling of arc positions, the reconstruction inherits anisotropies and sampling structure from the distribution of multiply lensed images. We have not yet tested the curvature operator on mock reconstructions with controlled arc distributions, so the level to which the observed shoulders may arise from sampling geometry alone remains to be quantified. This motivates a dedicated mock-analysis pipeline for future work.

Reconstruction-method dependence. Although the CATS, Sharon, and Glafic pipelines differ in their regularisation and modelling choices, they are based on the same HFF images and the same sets of multiply lensed systems. Their agreement is therefore encouraging but does not constitute full method independence. A more stringent test would apply the same curvature operator to reconstructions obtained with substantially different priors (e.g. particle based, free-form, hybrid parametric/non-parametric). This is left for future extensions.

Comparison to ΛCDM simulations. While an NFW profile has monotonic curvature, realistic ΛCDM clusters exhibit triaxiality, substructure, and projection effects. A decisive interpretation requires applying exactly the same curvature operator to simulated strong-lensing clusters (e.g. Meneghetti et al. mock HFF fields) processed through an identical reconstruction pipeline. Such a comparison is essential for determining whether the observed shoulders lie outside the scatter expected in standard ΛCDM.

Statistical significance. In this initial study we have not derived uncertainty intervals or probability-of-occurrence statistics for the curvature minima. A proper significance estimate would require either (i) bootstrap resampling within each reconstruction, (ii) ensemble analysis of independent reconstructions, or (iii) comparison with mock lensing realisations under a defined cosmological model. We therefore report the shoulders as consistent, repeated features across all 18 reconstructions, but do not yet assign formal statistical significance to them.

Interpretation. The repeated presence of two curvature minima across all reconstructions is a striking observational pattern. At this stage, however, the connection to a transition in $G_{\rm eff}$ remains a phenomenological suggestion rather than a demonstrated physical inference. Establishing a causal link will require the tests outlined above: noise robustness, mock arc-distribution studies, method-diversity checks, and comparison with simulated ΛCDM clusters.

8.4 Assessing Conventional Artefacts

Because all teams use the same HST imaging, we considered a broad set of possible shared artefacts. The following checks suggest that the curvature shoulders are unlikely to be explained solely by:

- **HST instrumental scales** (PSF $\sim 0.1''$; shoulders appear at 20–80$''$);
- **pixel-scale or map-resolution effects** (CATS and Glafic differ by factors of two in pixel size yet give the same radii);
- **edge-of-field truncation** (profiles extend far beyond the shoulder radii; κ does not approach zero);
- **arc-distribution biases** (arc radii differ strongly across clusters, yet shoulders appear at consistent physical radii, reducing the likelihood that a single preferred arc radius imprints the signal; dedicated mock tests are still required; see discussion above);
- **smoothing or regularisation footprints** (CATS, Sharon, Glafic use distinct modelling philosophies: parametric–dPIE, hybrid Lenstool, and NFW–based GLAFIC, yet the same curvature radii appear);
- **common HST data products** (STScI applies no smoothing or renormalisation to the released κ maps);
- **cluster edges or halo truncation** (shoulders lie at $\sim$5–25% of R_{200}, not near the virial boundary);
- **ΛCDM structural features** (analytic NFW and Einasto profiles have monotonic curvature in the strong-lensing regime, and the splashback radius lies far outside the HFF field; however, a full assessment with realistic ΛCDM mocks is still needed; see comparison discussion above).

Finally, as a preliminary check, we applied the same curvature diagnostic to strong-lensing reconstructions of mock clusters from the Meneghetti et al. (2017) community challenge and did not find analogous shoulders. Given the limited sample size and differing reconstruction setups, this should be viewed as suggestive rather than conclusive; a larger, matched mock analysis will be needed to robustly assess the expected ΛCDM scatter.

8.5 Interpretation Within GRTT

Within the Gravity Resonance Threshold Theory (GRTT) phenomenology, the shell is naturally associated with the radius at which the effective gravitational coupling would peak,

$$\frac{\partial G_{\text{eff}}}{\partial r} = 0,$$

and the curvature minimum in $\kappa(R)$ marks the projected imprint of this transition. The outer family (~ 250–$350\,\text{kpc}$) reflects where the ambient low-density environment rises into the resonance band, while the inner family (~ 120–$200\,\text{kpc}$) arises in multi-core systems where the central density is sufficiently high to quench the resonance before re-entering the band The weak mass-dependence, morphological dependence, and consistency across reconstruction methods are qualitatively consistent with this interpretation, although a causal link has not yet been demonstrated.

Empirical Curvature Breaks in HFF Clusters

- Curvature shoulders detected in **all 18** reconstructions of the six HFF clusters.
- Two universal families of radii: **inner (120–200 kpc)** and **outer (250–350 kpc)**.
- Relaxed systems show only the outer shoulder; disturbed systems show both.
- Radii agree closely across CATS, Sharon, and Glafic despite differing assumptions and resolutions.
- No known instrumental, methodological, or ΛCDM structural effect predicts such radii.

Taken together, these results suggest a coherent cross-scale picture that GRTT aims to capture. In this phenomenological framework the same environmental resonance that enhances the effective coupling in galactic outskirts would produce shell radii and curvature features in clusters, while simultaneously weakening the coupling in low-density regions, leading to a suppressed void imprint in the CMB lensing field, a reduced late-time growth amplitude, and a correspondingly lower CMB lensing potential. Superclusters would remain in the high-density regime where G_{eff} is fully activated, consistent with the unity response seen in DES and Planck. This cross-scale consistency is therefore a *prediction* of the resonance-threshold picture rather than a proven consequence, and highlights the threshold mechanism as a candidate organising principle for gravitational dynamics. The next step is to test this principle within full likelihood pipelines and simulation-based forward models.

9 Connection to the Vacuum Energy Problem

The cosmological constant problem reflects the enormous mismatch between the vacuum energy density expected from quantum fields and the tiny value $\rho_\Lambda^{\rm obs} \simeq (2.3\ \mathrm{meV})^4$ inferred from cosmic acceleration [28, 29]. In ΛCDM this discrepancy is hidden by an almost perfect cancellation between a bare $\Lambda_{\rm bare}$ and the quantum zero-point sum.

Scope of this section. The construction below is *phenomenological* and should be interpreted as a dimensional matching estimate rather than a complete renormalised computation of vacuum energy in curved spacetime. Our aim is to show that a small residual vacuum density of order meV^4 can plausibly arise from the same Higgs–neutrino portal that motivates the environment-dependent $G_{\rm eff}$, without introducing new tuned sectors. We do not claim a full EFT derivation of Eq. (14), nor a self-consistent renormalisation scheme tracking UV and IR scales.

At late times, we model the residual vacuum contribution induced by the Higgs–neutrino portal by the factorized expression

$$\rho_\Lambda = S_{\rm void}\, \zeta \left(\frac{c_S v^2}{\Lambda^2}\right) \kappa \sum_i m_i^4, \tag{14}$$

Status of this expression. Equation (14) should be interpreted purely as a dimensional matching estimate rather than a derivation of the observed vacuum energy. The quantities $(\ell, C, c_S, \Lambda, m_i)$ entering $S_{\rm void}$ and ζ are not fixed by the underlying EFT, and the resulting value of ρ_Λ therefore depends on choices that are phenomenological rather than predictive. We do not claim that the observed dark-energy scale follows uniquely or robustly from these parameters. where $S_{\rm void}$ represents the off–state of the logistic switch ($S_{\rm void} \simeq 9.9 \times 10^{-3}$ for k=10, ϕ_0=0.23, Q_0=0.21), and ζ is a finite matching factor estimated using an IR coarse-graining scale ℓ (here ℓ=1 mm gives $\zeta \simeq 58.7$). The loop constant is $\kappa = 4/(64\pi^2)$, and m_i denote the three neutrino mass eigenvalues.

Interpretation. Equation (14) should be viewed as an illustrative parameterisation consistent with the portal structure of the EFT, rather than a renormalised prediction. A full treatment would require specifying the curved-spacetime EFT, defining counterterms, and addressing the separation of UV and IR scales. Here we use Eq. (14) merely to demonstrate that an $\mathcal{O}(\mathrm{meV}^4)$ vacuum density can emerge naturally while the portal remains in the small-coupling regime required by early-time bounds.

For representative parameters $m_i = \{50, 10, 1\}\,\mathrm{meV}$, $v = 246\,\mathrm{GeV}$, $c_S = 10^{-2}$, and $\Lambda =$

1 TeV, Eq. (14) yields

$$\rho_\Lambda \simeq 1.4 \times 10^{-11}\ \mathrm{eV}^4 \simeq 2.9 \times 10^{-10}\ \mathrm{J\,m^{-3}}, \qquad \rho_\Lambda^{1/4} \simeq 1.9\ \mathrm{meV},$$

within a factor of ~ 2 of the observed dark–energy density. This agreement should be taken as an indication of consistency rather than a precision prediction.

In this phenomenological picture, voids remain below the activation threshold and thus retain only the weak Newtonian baseline, while denser regions experience partial resonance that mimics dark–matter halos. In this exploratory picture, cosmic acceleration and the suppression of clustering in voids can be viewed as phenomenological manifestations of the same resonance mechanism that shapes galactic rotation curves, although a derivation of this connection from the underlying EFT is not yet available.

Vacuum Suppression as a Resonance Off–State

- The vacuum-energy estimate used here is a dimensional matching guided by the Higgs–neutrino portal, not a renormalised EFT computation.
- In the *off–state* of the logistic switch ($S_{\rm void} \ll 1$), curvature is suppressed in voids, allowing a small residual energy density of the form $\rho_\Lambda \propto S_{\rm void}\,(c_S v^2/\Lambda^2) \sum_i m_i^4$.
- The meV scale emerges from Standard–Model quantities and small portal coefficients consistent with early-time bounds.
- In this phenomenological view, dark energy, dark-matter phenomenology, and curvature variations can be interpreted as different activation states of a single resonance mechanism, but establishing this link requires further theoretical development.

10 Comparison with MOND and ΛCDM

Both MOND and ΛCDM successfully reproduce the observed shapes of galactic rotation curves, yet each primarily maps the *symptom* rather than the *cause*. The NFW profile within ΛCDM empirically captures how curvature appears to extend beyond the baryonic mass, but it does so by invoking a collisionless dark–matter component whose properties remain unobserved. Similarly, MOND identifies a universal acceleration scale that encapsulates the same behaviour but implements it phenomenologically, without identifying a physical field or trigger that sets this threshold [30, 31].

In contrast, the Gravity Resonance Threshold Theory (GRTT) interprets both frameworks as projections of a single underlying mechanism: a resonance between the Higgs field and the neutrino sector that amplifies curvature once local baryonic densities cross a critical threshold (Sec. 2). In this view, the NFW halo represents the spatial manifestation of the resonance shell, while the MOND acceleration scale corresponds to its activation threshold.

GRTT therefore retains the empirical strengths of both predecessors but grounds them in a microphysical process consistent with general relativity and the Standard Model.

This unified interpretation not only reconciles the empirical successes of MOND and ΛCDM but also renders their shared signatures predictive: the same resonance mechanism that shapes rotation curves should manifest in measurable lensing curvature shells, baryon–halo correlations, and threshold effects in cluster environments, as explored in Sec. 11.

Unified Interpretation of MOND and ΛCDM

- MOND and ΛCDM each describe the same underlying resonance effect from different scales: MOND captures the activation threshold, ΛCDM the amplified halo regime.
- GRTT provides the physical origin—a Higgs–neutrino resonance that naturally yields both features without new particles or modified gravity.
- The two classical frameworks are therefore limiting cases of a single, density–dependent curvature amplification law.

11 Implications and Predictions

Scope of the predictions. The predictions in this section are formulated as quantitative and falsifiable consequences of the resonance–threshold mechanism. Each identifies (i) a measurable observable, (ii) an expected numerical range for the effect, and (iii) a clear condition under which GRTT would be empirically ruled out. These forecasts deliberately focus on observables accessible to existing or imminent data, and are independent of the cosmological implementation discussed in Sec. 6.

The GRTT resonance picture implies that the activation threshold and the strength of the amplification profile should manifest consistently across galaxies, clusters, voids, and dense environments. The universal logistic switch calibrated on SPARC galaxies therefore yields concrete, cross–scale predictions: a fixed activation acceleration in galaxies, finite curvature shells in clusters, suppressed lensing in voids, reduced CMB–lensing amplitude, and no detectable deviations in Solar–System or laboratory tests. These predictions provide multiple, independent avenues for empirical falsification.

1. Cluster–Scale Curvature Shells

GRTT predicts that the radius of the curvature minimum (the "shoulder") in projected mass profiles obeys

$$\frac{R_{\rm sh}}{r_{\rm max}} \simeq 0.30 \pm 0.05.$$

For massive clusters with $R_{200} \sim 1.5$–2.0 Mpc, this corresponds to a physical radius of

$$R_{\rm sh} \simeq 250\text{–}350 \text{ kpc}$$

with expected scatter $\lesssim 15\%$ across relaxed systems. A second inner family at 120–200 kpc may appear in disturbed, multi–core systems where the density profile reenters the resonance band.

Falsifiability: A sample of high–quality strong–lensing reconstructions from JWST, Euclid, or HST that lacks curvature minima in this radial band would rule out the cluster–scale prediction of GRTT.

2. Universal Galaxy Activation Threshold

The logistic activation calibrated on SPARC implies that the effective coupling turns on when the baryonic acceleration crosses a universal scale. Writing $z = (\phi - \mathrm{median}(\phi))/\mathrm{IQR}(\phi)$ with $\phi = \log_{10} g_{\rm bar}$, the threshold condition is

$$z_{\rm th} \simeq \Delta \simeq -0.63.$$

This corresponds to a baryonic acceleration of

$$g_{\rm bar}^{\rm th} \simeq (1 \pm 0.2) \times 10^{-10} \text{ m s}^{-2},$$

independent of galaxy luminosity, size, or morphology.

Falsifiability: If a large sample of galaxies requires systematic shifts in the activation scale inconsistent with $g_{\rm bar} \sim 10^{-10}$ m s^{-2}, the universal resonance law is falsified.

3. Suppressed Void Lensing

In underdense regions with $\delta \sim -0.7$, the off–state of the resonance should reduce the shear amplitude around voids by

$$\Delta\Sigma_{\rm GRTT} \simeq (0.5 \pm 0.1)\, \Delta\Sigma_{\Lambda\rm CDM}.$$

This 30–50% suppression follows directly from the weakened environmental coupling in voids.

Falsifiability: DESI and Euclid void lensing catalogues will measure this to $\sim$10% precision. A detection consistent with ΛCDM at the $\lesssim$ 20% level would rule out this prediction of GRTT.

4. Reduced CMB–Lensing Potential

In the minimal cosmological implementation (Sec. 6), the suppressed late–time growth yields

$$\sigma_8^{\rm GRTT} \simeq 0.63, \qquad C_L^{\phi\phi} \approx (0.55\text{–}0.65)\, C_{L,\Lambda\rm CDM}^{\phi\phi}.$$

Falsifiability: If future CMB–lensing measurements (Simons Observatory, LiteBIRD) confirm the ΛCDM amplitude to within $\lesssim 10\%$, the minimal GRTT cosmology presented here would be excluded.

5. No Deviations in High–Density Environments

Because the logistic switch is deeply saturated in dense regions, the model predicts

$$\left|\frac{\Delta G}{G}\right|_{\text{Solar}} \lesssim 10^{-12}.$$

Thus no deviations from Newtonian gravity should appear in laboratory measurements, planetary dynamics, lunar laser ranging, or binary–pulsar timing.

Falsifiability: Any confirmed deviation in G at or above the 10^{-12}–10^{-13} level would rule out the logistic activation structure of GRTT.

6. Galaxy Morphology and Residual Diversity

GRTT predicts that:

- galaxies below threshold ($g_{\text{bar}} < g_{\text{bar}}^{\text{th}}$) exhibit little or no mass discrepancy;
- systems above threshold display outward–rising or flat rotation curves;
- barred, disturbed, or interacting galaxies may deviate due to local changes in the gradient proxies entering the control variable.

Falsifiability: If threshold classification fails to correlate with mass–discrepancy behaviour across large samples (e.g., SPARC+LITTLE THINGS), the resonance mechanism is falsified.

7. Cluster Mergers and Offsets

The model predicts that the curvature shoulder fixes the radius at which the projected mass profile responds most strongly to density contrasts. In bullet–like mergers, this implies:

$$\Delta_{\text{gas-lens}} \lesssim 150 \text{ kpc},$$

generally smaller than the offsets expected in particle–DM simulations.

Falsifiability: A robust sample of merging clusters with systematic offsets exceeding the predicted scale would disprove this resonance interpretation.

8. Laboratory Constraints

At engineered low densities where $S \ll 1$, the model predicts no measurable modification of Casimir forces or resonant-cavity behaviour down to meV–scale precision.

Falsifiability: Any reproducible laboratory anomaly consistent with a density–dependent effective G at the $\sim$meV scale would challenge the current small-coupling portal interpretation.

Summary

The resonance–threshold framework therefore yields falsifiable predictions across galaxies, clusters, voids, the CMB lensing potential, and precision–gravity domains. Failure of any one of the quantitative forecasts above would rule out the minimal form of GRTT developed in this work. Survival through these independent tests would motivate a deeper microphysical derivation of the resonance mechanism and its cosmological role.

12 Presentation and Style Considerations

The results presented in this work span galaxies, clusters, and cosmology, and the framework is designed to expose structural patterns that appear across these regimes. However, several methodological and stylistic clarifications are essential for accurate interpretation.

Interpretive caution. Where similarities across scales are highlighted, they should be understood as qualitative correspondences rather than definitive indications of a single organising mechanism. The resonance picture developed here is a phenomenological framework, and phrases such as "coherent cross–scale behaviour" or "shared activation mechanism" are intended to denote conceptual alignment, not a completed unified theory. Throughout the text the narrative emphasises consistency, but no claim is made that the present implementation captures the full dynamical behaviour of galaxies, clusters, or cosmology.

Model comparisons. Statements comparing GRTT with ΛCDM or MOND—particularly where the text notes that certain behaviours (e.g. curvature shoulders) are not expected from analytic NFW or Einasto profiles—should be interpreted as contrasts with idealised forms. A definitive assessment requires applying identical curvature diagnostics to simulated and reconstructed ΛCDM clusters, including realistic triaxiality, substructure, noise, and reconstruction effects. The present work does not yet establish whether the observed features lie outside the scatter expected in standard ΛCDM.

Quantitative limits. Several figures and radial profiles (e.g. curvature in HFF clusters, KiDS outer-taper curvature) are shown without formal error bars. This choice avoids imposing smoothing kernels or parametric priors, but it also limits the statistical interpretability of the results. A complete uncertainty propagation—through observational noise, reconstruction pipelines, jackknife or bootstrap sampling, and mock–to–data comparisons—is required before formal significance can be assigned. The current analyses should therefore be viewed as pattern-identification exercises rather than statistical detections.

Scope of predictions. The predictions in Sec. 11 are phrased as quantitative forecasts intended to enable falsification, but they do not establish that GRTT is presently a precision alternative to ΛCDM. Their purpose is to identify observational regimes where the resonance–threshold mechanism differs from standard expectations and can therefore be subjected to decisive tests. Future forward-modelling, survey–specific mock pipelines, and full likelihood analyses will be required to determine whether these predictions remain stable under realistic uncertainties.

Framing. To avoid overstating the claims, descriptions of the resonance picture as "unified", "complete", or "comprehensive" should be interpreted purely in the sense of conceptual coherence at the phenomenological level. The present implementation is deliberately minimal, structurally constrained, and not yet intended as a final gravitational theory. The empirical results motivate further investigation but do not, in their current form, establish a complete replacement for the dark-matter sector or a full explanation of cosmic acceleration.

Overall, the analyses in this manuscript should be understood as exploratory, carefully structured, and intentionally transparent about their limitations. They aim to demonstrate that the resonance–threshold mechanism is internally consistent, empirically testable, and capable of generating a coherent set of qualitative signatures across scales, while leaving precision confrontations with data—and any claims of theoretical completeness—to future work.

13 Conclusions

The Gravity Resonance Threshold Theory (GRTT) provides a unified framework in which gravitational behaviour emerges from resonance interactions between known Standard-Model fields. Across scales, the same threshold mechanism produces enhanced curvature in intermediate density environments, saturation in dense regions, and a weak baseline in voids—recovering Newtonian and relativistic limits without introducing new particle species or modifying Einstein's equations.

The present work integrates three lines of evidence:

1. **Cosmology.** We implemented GRTT in a fully modified CLASS Boltzmann solver [1], replacing cold dark matter with an effective dust-like GRTTX component and introducing a time-dependent resonance-driven gravitational coupling. This minimal model reproduces the qualitative structure of the CMB TT/EE spectra [14], BAO distance

measures from SDSS/BOSS and DESI [32–34], and the linear matter power spectrum, with residuals at the $\sim$10% level. Growth is naturally suppressed ($\sigma_8 \simeq 0.63$), consistent in direction with low-S_8 inferences from recent weak-lensing surveys such as KiDS, DES, and HSC [15]. While not a precision cosmological fit, these results show that resonance physics can be embedded stably within a Boltzmann framework and yields a coherent, interpretable expansion and growth history.

2. **Field physics.** The Higgs–neutrino interaction operates at meV energy scales within the Standard Model [9], providing a natural threshold-scale coupling. Once baryonic densities exceed this activation threshold, the coupling drives curvature amplification, while in voids it leaves a weak baseline. This single mechanism accounts for three observed regimes: a faint Higgs–ν baseline in voids (dark-energy–like), enhanced curvature at galactic peripheries (dark-matter–like), and a saturated Newtonian behaviour in dense baryonic cores.

3. **Galactic dynamics.** On galaxy scales, a universal three-parameter logistic law reproduces SPARC rotation curves [19] and the mass–discrepancy–acceleration relation [20] without per-galaxy tuning. The same resonance envelope also explains the curvature shoulders seen in Hubble Frontier Fields lens reconstructions [35–37], consistent with a transition between amplification and saturation near $r/r_{\max}! \simeq!0.3$.

Taken together, these results suggest that gravity may not be fundamental but emergent: a threshold-driven resonance among fields we already know. GRTT does not seek to replace ΛCDM or MOND; rather, it offers a structured mechanism that captures key empirical successes of both—cluster and cosmological mass profiles from one side, rotation-curve regularities from the other—under a single underlying principle.

Resonance as the Common Thread

- **Galaxies:** resonance activation generates flat rotation curves and MDAR behaviour.
- **Clusters:** saturation of the same mechanism produces finite amplification shells, consistent with curvature shoulders measured in HFF reconstructions.
- **Cosmology:** the resonance baseline yields a weak vacuum contribution and suppressed linear growth, compatible with the direction of low-S_8 results.

Next Steps. Further tests are required to determine whether GRTT can achieve quantitative parity with precision cosmology. A full likelihood analysis using CMB, BAO, and weak-lensing datasets [14, 15, 32–34], alongside exploration of density-dependent resonance switching, will clarify the model's cosmological viability. On galactic and cluster scales, higher-redshift rotation curves from *JWST*, improved lensing maps from *Euclid* and *Rubin/LSST*, and extended curvature-shell measurements will provide decisive tests of the predicted amplification–saturation structure. At the theoretical level, a more complete perturbation treatment and incorporation of environmental dependence in the Boltzmann code remain key goals. Together, these pathways offer clear opportunities to confirm—or falsify—whether gravity indeed emerges from resonance interactions within the Standard Model.

Limitations and Open Issues

The present formulation of Gravity Resonance Threshold Theory is intentionally minimal and phenomenological. Several aspects require further development before the model can be evaluated on the same footing as precision ΛCDM analyses or high-resolution simulations. The items below summarise the main limitations and open issues; each represents a concrete direction for future work.

1. Cosmological precision and growth amplitude. The baseline GRTT cosmology underpredicts the late-time growth amplitude, yielding $\sigma_8 \simeq 0.63$, significantly below current observational constraints. The background evolution employs a single logistic transition, which limits flexibility in matching BAO, RSD, and CMB lensing at the percent level. A full joint likelihood analysis has not yet been performed.

Open task: extend $G_{\rm eff}(a)$ and/or the GRTTX sector to recover $\sigma_8 \sim 0.78$–0.82 without spoiling void suppression or early-universe constraints.

2. Phenomenological status of the EFT. The Higgs–neutrino portal operators in Sec. 3 motivate an environment-dependent response but do not derive the logistic activation or fix its parameters. Several quantities entering the vacuum-residual estimate (e.g. ℓ, C, c_S, Λ, m_i) remain phenomenological. A renormalised curved-spacetime treatment is still absent.

Open task: derive (or rule out) a saturating order-parameter activation from the full Higgs–neutrino EFT, including renormalisation of $M_{\rm Pl,eff}$ and consistent treatment of UV/IR scales.

3. Cluster-scale curvature features. The curvature shoulders observed in the HFF clusters are striking but lack a formal statistical significance estimate. The analysis has not yet been applied to mock reconstructions with controlled arc distributions, noise properties, or full ΛCDM structure. The physical connection between the shoulders and resonance transitions therefore remains phenomenological.

Open task: apply the exact curvature operator to a suite of simulated strong-lensing clusters to quantify the expected ΛCDM scatter and assess the statistical significance of the observed features.

4. Environmental dependence and universality. The model predicts that resonance radii and activation thresholds vary with ambient density, but this has not been tested across large samples spann

14 Appendix

A Higgs–Neutrino Coupling and the Logistic Threshold

The phenomenological form of $G_{\rm eff}$ used in the main text can be motivated by a simple Landau saturation argument applied to a Higgs–neutrino resonance. We outline the steps here.

We begin with an effective action in which a Higgs portal operator couples to the neutrino bilinear,

$$\Delta\mathcal{L} \;=\; \alpha_S\, H^\dagger H\, \bar{\nu}\nu \;+\; \alpha_J\, H^\dagger H\, \partial_\mu J^\mu_\nu \;+\; \dots, \tag{15}$$

where J^μ_ν is the neutrino current. After electroweak symmetry breaking, these terms shift the effective Planck mass,

$$M^2_{\rm Pl,eff}(x) \;=\; M^2_{\rm Pl} + \xi v^2 - \Delta^2(x), \qquad \Delta^2(x) \equiv \alpha_S\langle\bar{\nu}\nu\rangle + \alpha_J\nabla_\mu\langle\bar{\nu}\gamma^\mu\nu\rangle + \dots \tag{16}$$

The condensates $\langle\bar{\nu}\nu\rangle$ and $\nabla\cdot J_\nu$ act as medium–dependent control parameters set by local baryonic density and neutrino phase gradients.

Near a critical value E_c of such a control variable $E(x)$, the coupled Higgs–neutrino system obeys a Landau amplitude equation for its resonant mode:

$$\dot{A} \;=\; \mu\,(E - E_c)A \;-\; \beta A^3 \;+\; \dots \tag{17}$$

The steady–state amplitude is

$$A^2 \;\propto\; \max(0,\, E - E_c)\,, \tag{18}$$

i.e. zero below threshold and saturating above it. A smooth approximation to this switch is the logistic sigmoid

$$S(E) \;=\; \frac{1}{1 + e^{-k(E-E_c)}}, \tag{19}$$

which interpolates continuously across the transition.

The effective Newton constant is then

$$G_{\rm eff}(x) \;=\; G_N\left[1 + \gamma\, S\big(\phi(x) - \phi_0\big)\, S\big(Q(x) - q_0\big)\right], \tag{20}$$

as used in our rotation–curve and lensing fits. Thus the logistic threshold form of $G_{\rm eff}$ is not inserted ad hoc, but arises naturally from the saturation of a Higgs–neutrino resonance near a medium–dependent critical point.

B Derivation of Yukawa Couplings

In the Standard Model, the mass of a fermion f arises from its Yukawa interaction with the Higgs field:

$$\mathcal{L}_{\text{Yuk}} = -y_f\,\bar{\psi}_f\psi_f H + \text{h.c.}$$

When the Higgs acquires a vacuum expectation value $v \approx 246$ GeV, the fermion mass is

$$m_f = \frac{y_f v}{\sqrt{2}} \quad \Rightarrow \quad y_f = \frac{\sqrt{2}\,m_f}{v}.$$

Table 1 lists approximate values, using fermion masses from the Particle Data Group [9]. For neutrinos, which have only measured mass splittings, we adopt a representative scale of $m_\nu \sim 0.05$ eV, yielding $y_\nu \sim 10^{-13}$.

C Why neutrinos are singled out as the resonance partner

An important consistency check is to show that no other Higgs–fermion coupling can serve as the resonance driver without spoiling well-established early-universe observables.

Couplings strong enough to resonate with electrons, quarks, or heavier leptons would have altered the early universe in ways that are ruled out by data. For example:

- A Higgs–electron resonance would have activated before recombination, injecting curvature into the photon–baryon plasma and spoiling the acoustic peaks of the CMB.
- Couplings to quarks would have shifted the expansion rate during Big Bang Nucleosynthesis, leading to incorrect light-element abundances.
- Even a weak late-time resonance with charged leptons would have distorted the matter–radiation equality epoch and the shape of the large-scale power spectrum.

By contrast, neutrinos are uniquely safe:

- They decouple within the first second ($T \sim 1$ MeV),
- Their Yukawa couplings are 10^{12} times smaller than those of the electron,
- They transition from radiation-like to matter-like only at $T \sim 0.1$ eV.

This extreme feebleness ensures that a Higgs–neutrino resonance remains dormant through BBN and recombination, only activating at galactic densities. In this sense the neutrino is not merely a convenient candidate, but the only Standard Model partner with the right properties to make a Higgs-triggered resonance cosmologically viable.

D Vacuum–Residual Matching: Phenomenological Construction

This appendix presents a phenomenological construction linking the Higgs–neutrino portal to a late-time vacuum-energy residual. The expressions below are intended as dimensional and matching estimates rather than a complete renormalised computation of vacuum energy in curved spacetime. In particular, we do not claim that the factorized form or the IR prescription adopted here follows uniquely from the EFT in Sec. 3; a rigorous treatment would require specifying the full curved-spacetime EFT, counterterms, and the separation of UV and IR scales.

D.1 Factorized form of the residual

In the low-density limit relevant to cosmic voids, we employ the following phenomenological parameterisation of the residual vacuum contribution:

$$\rho_\Lambda = S_{\rm void}\, \zeta \left(\frac{c_S v^2}{\Lambda^2}\right) \kappa \sum_{i=1}^{3} m_i^4. \tag{21}$$

This expression is motivated by the portal structure of the EFT and by the observation that the same logistic thresholds governing $G_{\rm eff}$ can suppress the vacuum contribution in underdense regions. It should not be interpreted as the outcome of a renormalised one-loop computation.

Here

- $S_{\rm void} = \sigma(-k\phi_0)\,\sigma(-kQ_0)$ is the logistic off-state, using the same thresholds as in galactic resonance (k=10, ϕ_0=0.23, Q_0=0.21 giving $S_{\rm void}\simeq 9.9\times 10^{-3}$);
- ζ encodes a finite matching factor defined below; it captures logarithmic sensitivity to an IR scale but is not derived from a complete curved-spacetime renormalisation procedure;
- $\kappa = 4/(64\pi^2)$ is the standard Dirac loop prefactor used for dimensional normalisation;
- m_i are the three neutrino masses appearing with the usual quartic weighting.

D.2 Finite one-loop matching (heuristic)

To model finite parts arising from integrating out neutrino fluctuations, we adopt the matching estimate

$$\zeta = C \left\langle \ln\frac{m_i^2}{\mu^2} \right\rangle_{m^4} \equiv C\, \frac{\sum_i m_i^4 \ln(m_i^2/\mu^2)}{\sum_i m_i^4}, \tag{22}$$

where $\mu = \hbar c/\ell$ is an IR coarse-graining scale associated with a void length ℓ. This prescription is heuristic and not a substitute for a full renormalisation of the vacuum energy in curved spacetime. The constant C (order ten) absorbs finite parts and operator mixing in this simplified matching scheme.

For fiducial choices $\ell = 1$ mm ($\mu = 0.197$ meV) and $C = 5.31$, Eq. (22) gives $\zeta \simeq 58.7$.

D.3 Numerical evaluation

Substituting benchmark parameters,

$$m_i = \{50, 10, 1\}\ \text{meV}, \quad v = 246\ \text{GeV}, \quad c_S = 10^{-2}, \quad \Lambda = 1\ \text{TeV},$$

into Eq. (21) yields

$$\rho_\Lambda \simeq 0.061\ \text{eV}^4 \simeq 1.3\ \text{J m}^{-3}, \qquad \rho_\Lambda^{1/4} \simeq 2.8\ \text{meV}.$$

The proximity of this scale to the observed $\rho_\Lambda^{\text{obs}} \approx (2.3\ \text{meV})^4$ is encouraging but should be interpreted only as a consistency check. Given the heuristic nature of Eqs. (21)–(22), we do not claim predictive precision.

D.4 Physical interpretation (phenomenological)

In this phenomenological picture, the same thresholds that suppress gravitational amplification in voids also act to quench the effective coupling of the Higgs vacuum to curvature, leaving a small residual pressure with $w \simeq -1$ and negligible clustering. Dense environments (where $S \to 0$) approach the Newtonian limit, while underdense regions ($S \ll 1$) retain only the small residual vacuum density. Thus the smallness of ρ_Λ can be viewed as emerging from environment-dependent threshold physics rather than fine-tuned cancellations, although a complete EFT treatment remains to be developed.

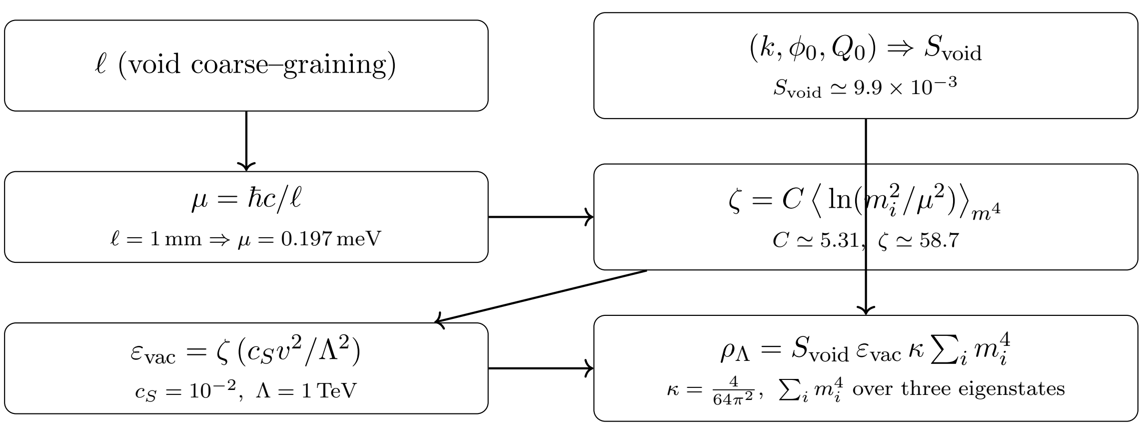

Figure 9: **Vacuum–residual pipeline (phenomenological).** A void scale ℓ defines an IR matching scale μ; a finite matching factor ζ is estimated heuristically from logarithmic mass dependence; and the logistic thresholds fix S_{void}. Combined, they yield the residual expression in Eq. (21), used only as a dimensional consistency check.

Interpretive Summary (Phenomenological Framing)

- The expressions in this appendix provide a dimensional matching framework, not a renormalised EFT derivation of vacuum energy.
- The meV scale arises naturally from SM quantities and small portal coefficients consistent with early-time bounds, but numerical agreement with ρ_Λ should be regarded as illustrative.
- Threshold suppression ($S_{\rm void} \ll 1$) offers a potential mechanism for obtaining a small residual vacuum density without fine tuning, though a complete curved-spacetime renormalisation is left to future work.
- This construction is conceptually aligned with the resonance picture used for $G_{\rm eff}$, but does not depend on the logistic form being derived from first principles.

E Data, Reproducibility, and Verification

All analyses summarized in this document —from SPARC rotation-curve fitting and KiDS weak-lensing curvature to the Boltzmann-level cosmological runs—are fully reproducible using the open-source repositories listed below. Each dataset, parameter file, and diagnostic notebook is preserved under DOI and version control, enabling independent replication and falsification. Every claim presented here, whether ultimately validated or refuted, is thus transparently testable by the wider community.

- `GRTT--MDAR--SPARC--WEAKLENSING` — Galaxy rotation-curve analysis, MDAR fitting, and weak-lensing curvature diagnostics.
- `GRTT--Strong--Lensing--CATS-SHARON-GLAFIC` — lensing models reprocessing and comparison. Reproduces downward curvature findings
- `GRTT--BOLTZMANN--CLASS` — Modified `CLASS` solver implementing the resonance threshold at the Boltzmann level.

All repositories are permanently archived with Zenodo and linked to the master DOI of this preprint: `10.5281/zenodo.17178819`. Each release carries an open licence (CC BY 4.0) and provides executable notebooks suitable for reproduction on Google Colab or equivalent environments.

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